

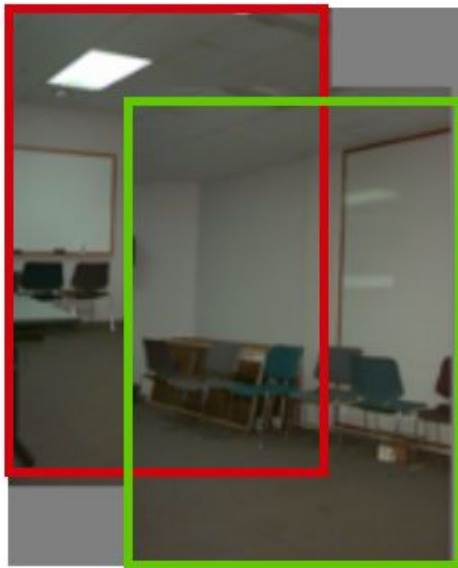
Computer vision

3D reconstruction

Geometric Transformation -recap.

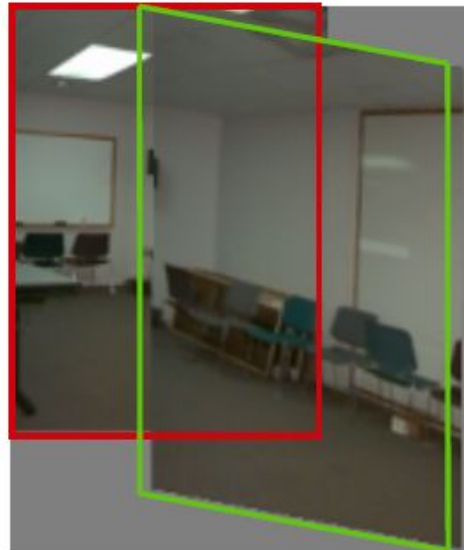
Translation

2 unknowns (2×1)



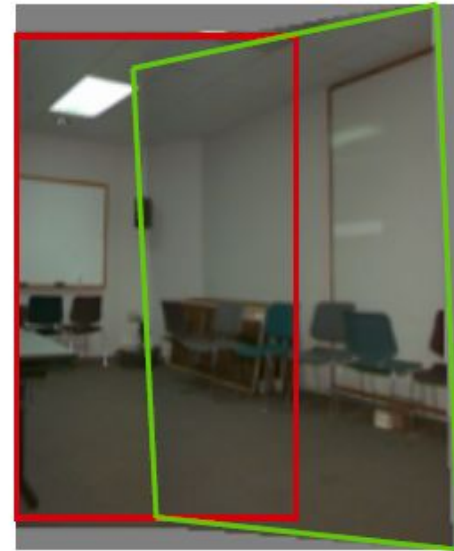
Affine trans.

6 unknowns (2×3)



Projective trans.

8 unknowns (2×4)



2D Affine Transformations

...sets of parallel lines remain parallel after an affine transformation...

2D Affine Transformations

An *affine transformation* can be written in the form:

$$x' = a_{xx}x + a_{xy}y + b_x$$

$$y' = a_{yx}x + a_{yy}y + b_y$$

or

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_{xx} & a_{xy} \\ a_{yx} & a_{yy} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \end{bmatrix}$$

=> linear transformation followed by a translation

Composition of Affine Transforms

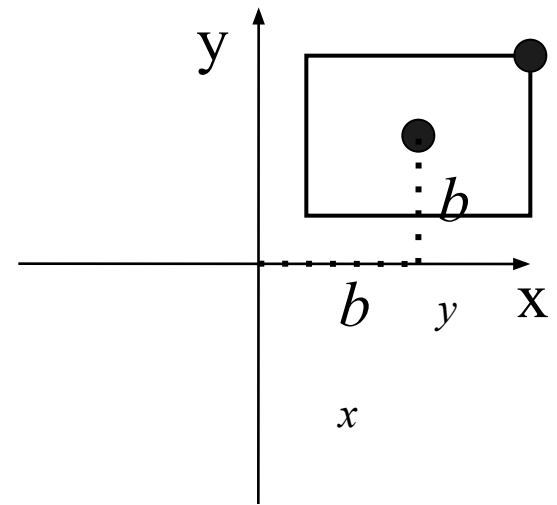
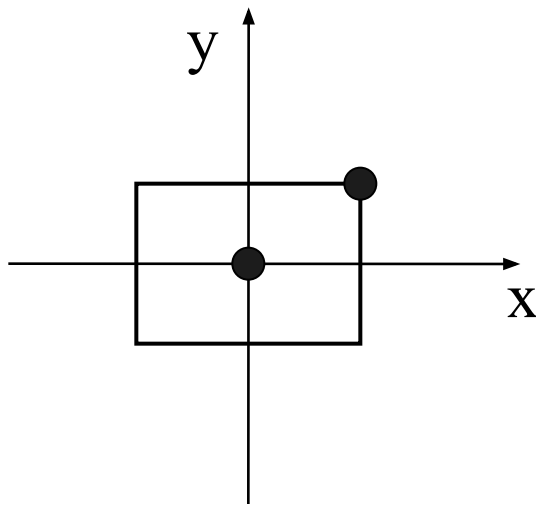
Any affine transformation can be composed as a sequence of simple transformations:

- Translation
- Scaling
- Rotation
- Shear (deformation)
- Reflection

2D Translation

Moves an object

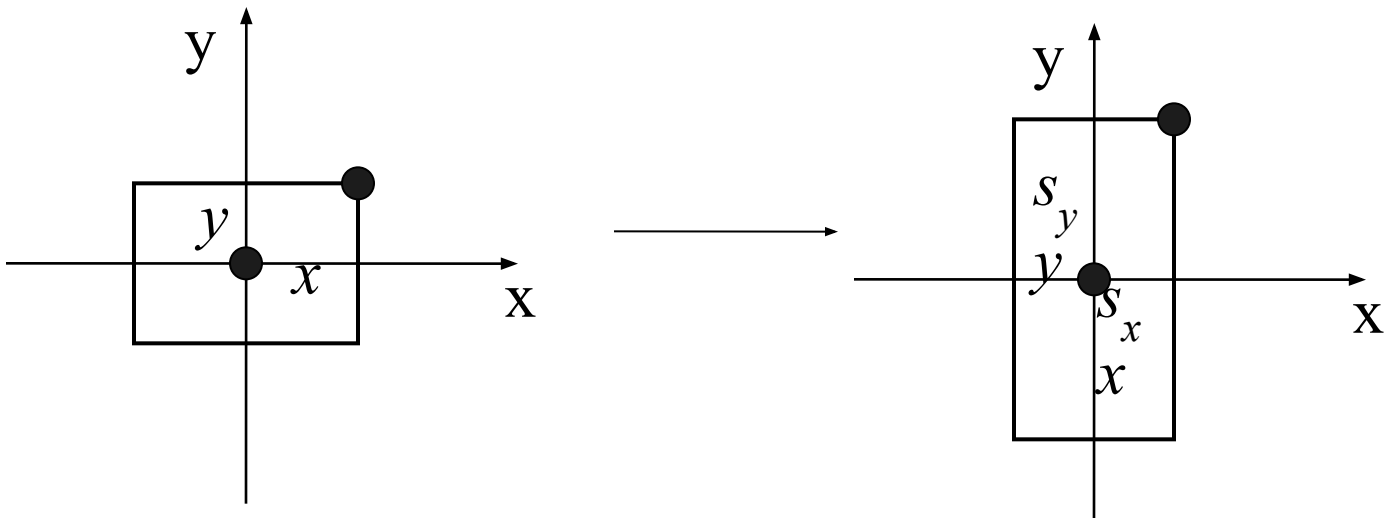
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \end{bmatrix}$$



2D Scaling

Resizes an object in each dimension

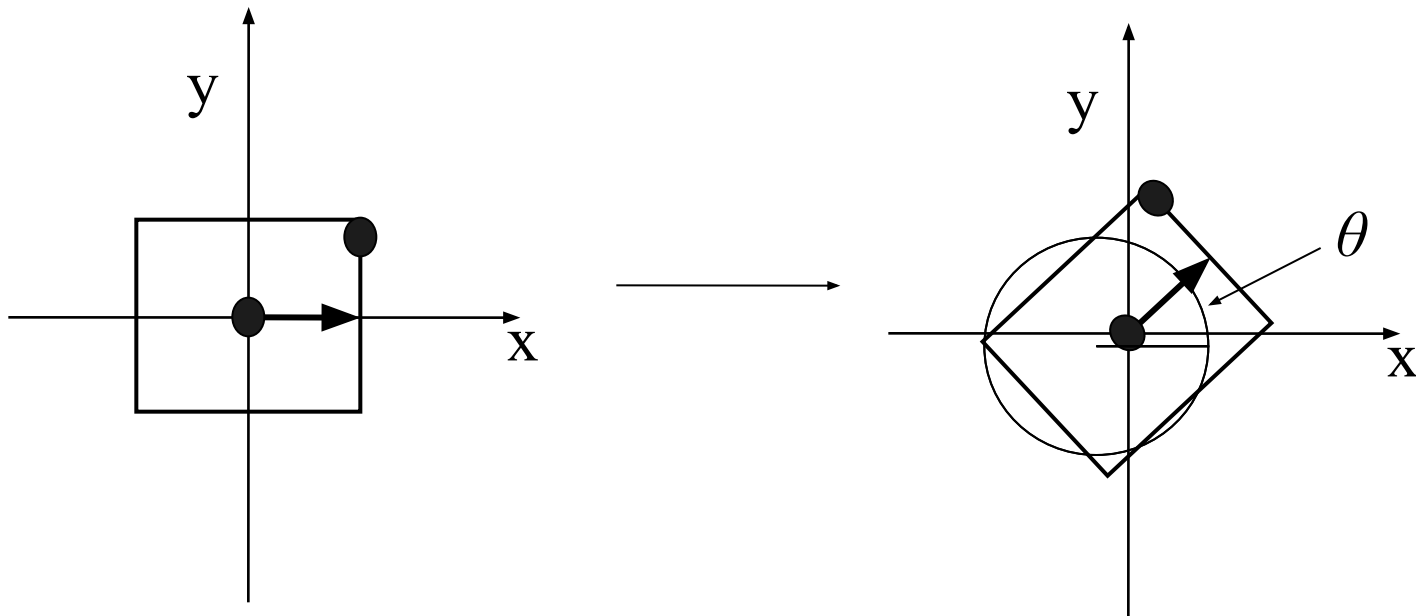
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



2D Rotation

Rotate counter-clockwise about the origin by an angle θ

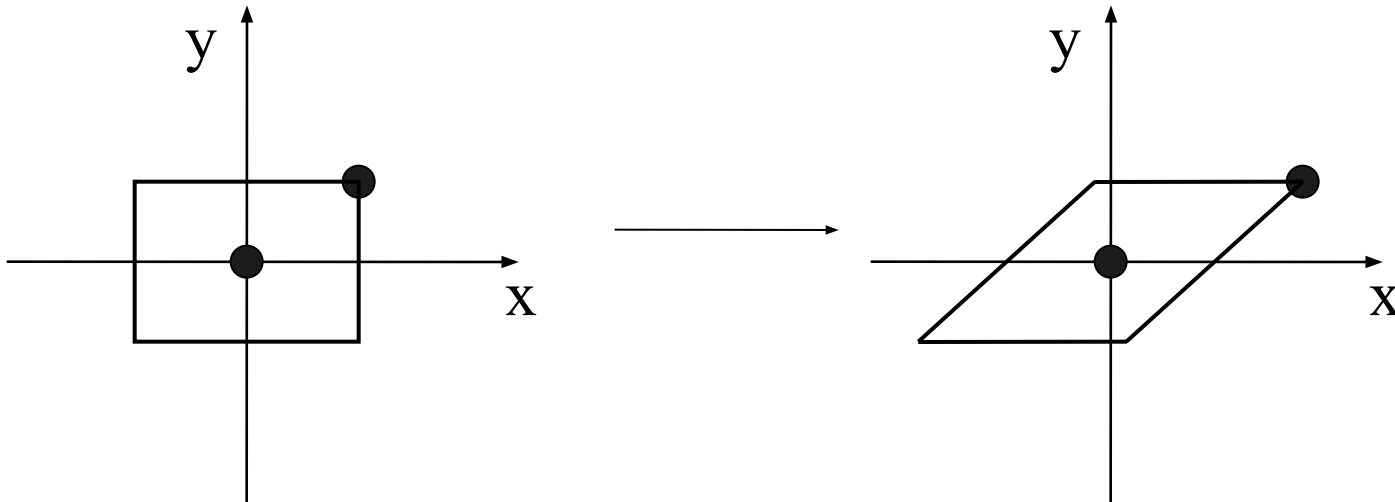
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



X-Axis Shear

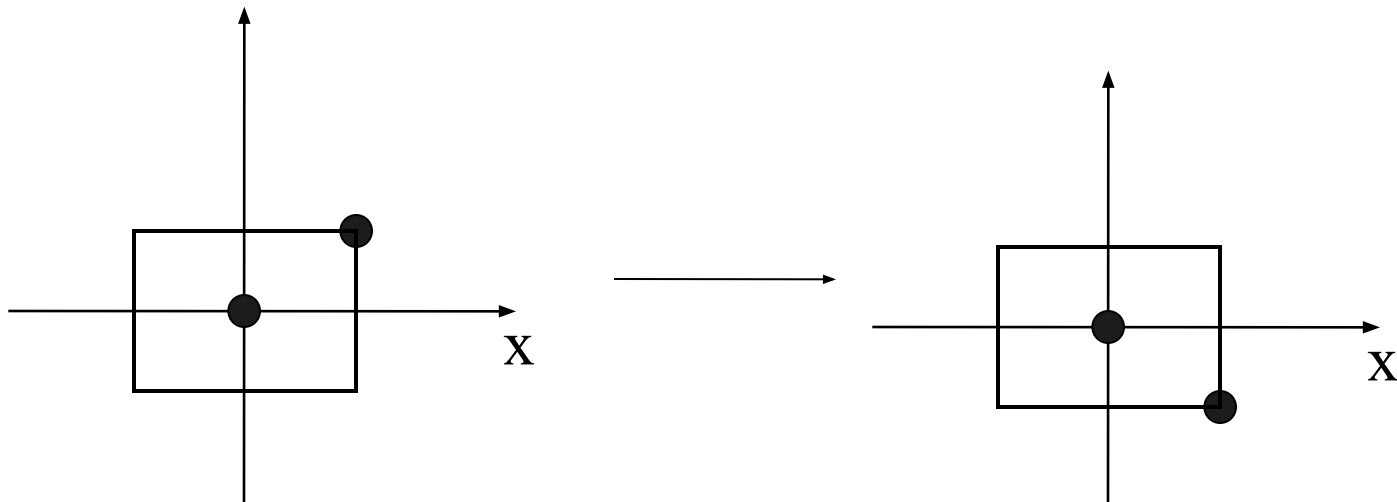
Shear along x axis

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Reflect About X Axis

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



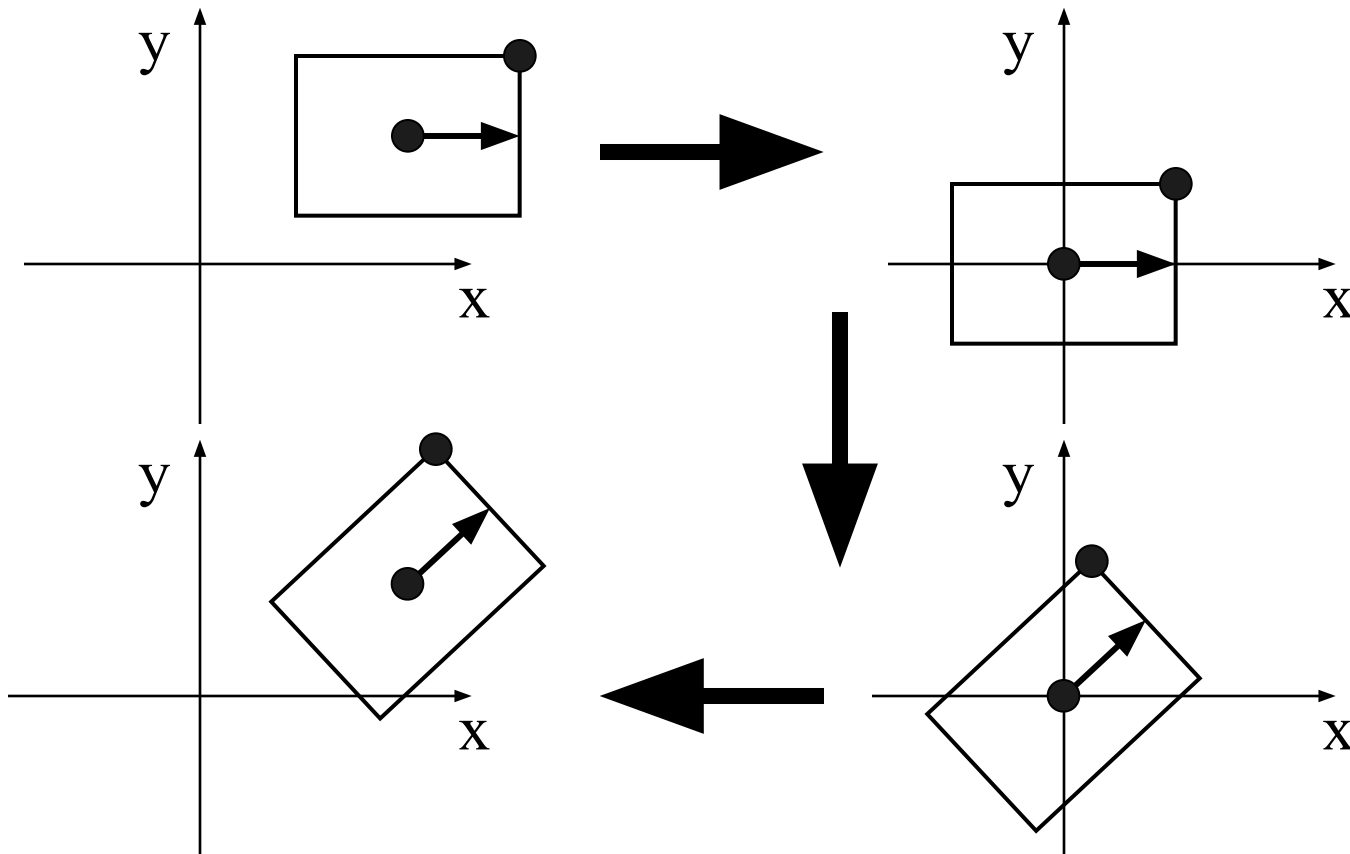
Rotating About An Arbitrary Point

What happens when you apply a rotation transformation to an object that is not at the origin?

Solution:

- Translate the center of rotation to the origin
- Rotate the object
- Translate back to the original location

Rotating About An Arbitrary Point



Homogeneous Coordinates

Use three numbers to represent a point

$(x,y)=(wx, wy, w)$ for any constant $w \neq 0$

Typically, (x,y) becomes $(x,y,1)$

Translation can now be done with matrix multiplication!

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a_{xx} & a_{xy} & b_x \\ a_{yx} & a_{yy} & b_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D Basic Transformations in Homogeneous Coordinates

Translation:

$$\begin{bmatrix} 1 & 0 & b_x \\ 0 & 1 & b_y \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation:

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Scaling:

$$\begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

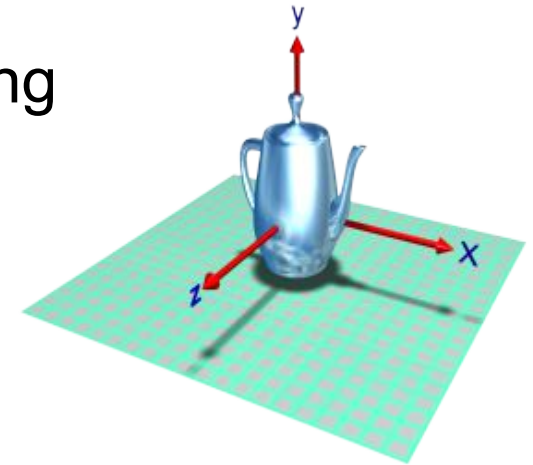
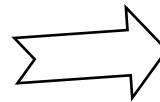
Homogeneous Transform Advantages

- Unified view of transformation as matrix multiplication
Easier in hardware and software
- To compose transformations, simply multiply matrices
- Allows for non-affine transformations:
Perspective projections!

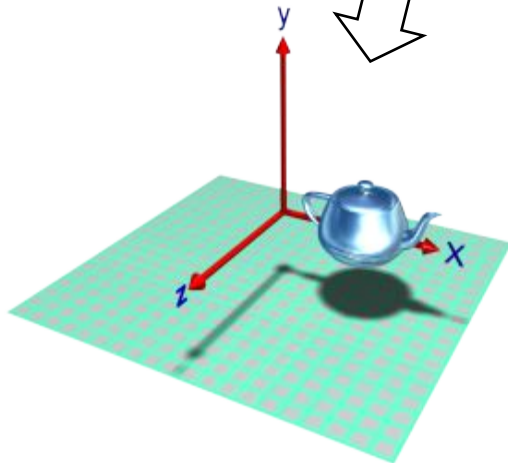
3D Affine Transformations

3D Object Transformations

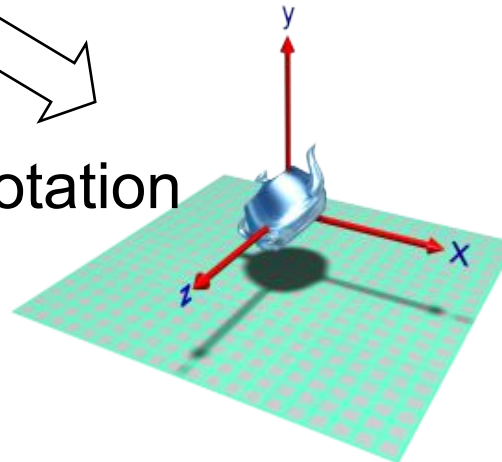
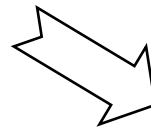
scaling



translation

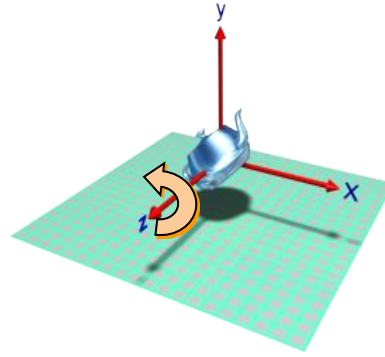
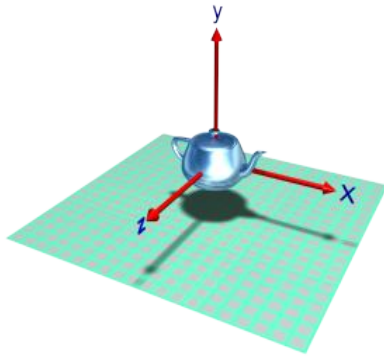


rotation

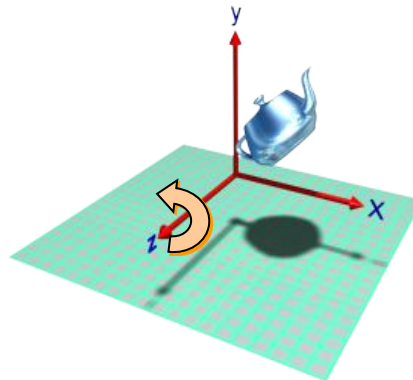
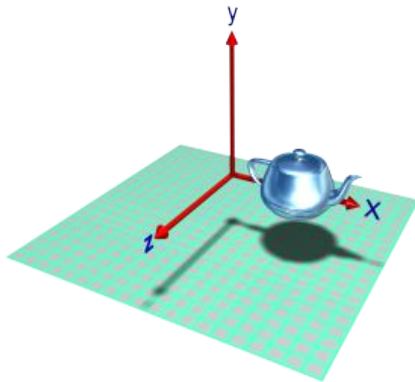


Rotation

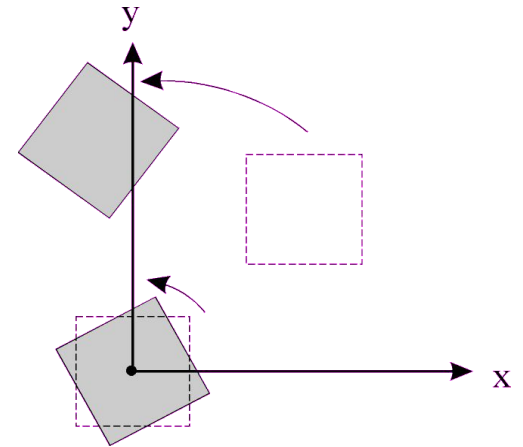
anti-clockwise about the origin:



rotation of 45° about the Z axis



offset from origin rotation



3D Transforms in Homogeneous Coordinates

Most of the time when working with 3D, we are thinking in terms of Euclidean geometry – that is, coordinates in 3- dimensional space (X, Y, Z)

Projective geometry has an extra dimension, called W , in addition to the X , Y , and Z dimensions. (X, Y, Z, W)

This 4-dimensional space is called "projective space," and coordinates in projective space are called "homogeneous coordinates".

3D Basic Transformations in Homogeneous Coordinates

⇒ 3D homogeneous coordinates: 4 dimensions

Translation:

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

Scaling:

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

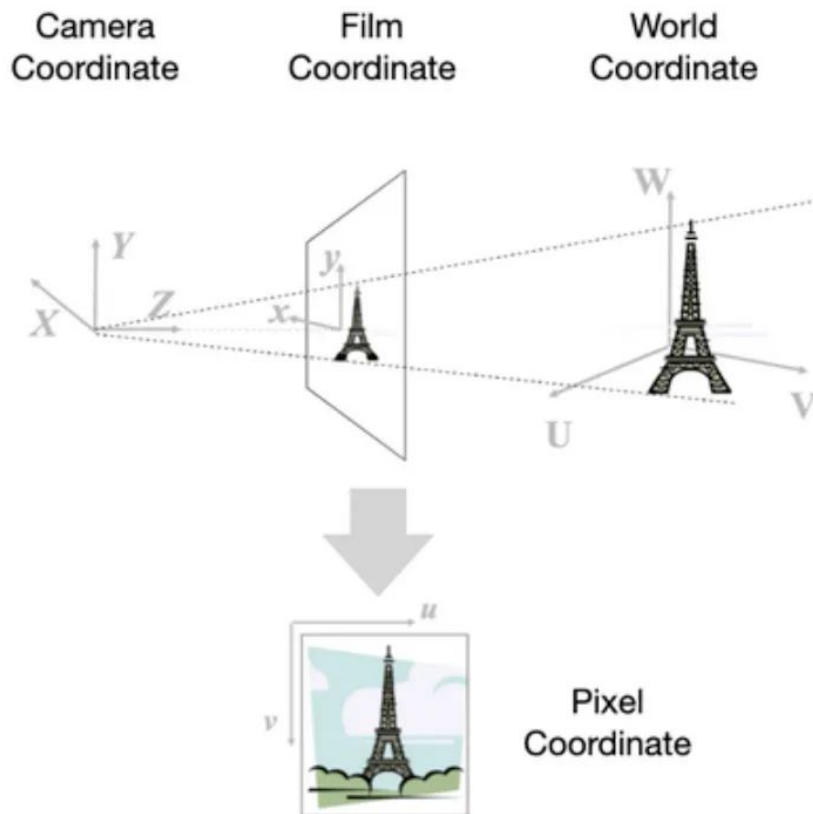
3D Basic Transformations in Homogeneous Coordinates

3D homogeneous rotations:

$$\mathbf{R}_x = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \mathbf{R}_y = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$\mathbf{R}_z = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

World coordinate to image coordinate

... world coordinate and camera coordinate have 3 dimensions while the film coordinate and image pixel coordinate have 2 dimensions.

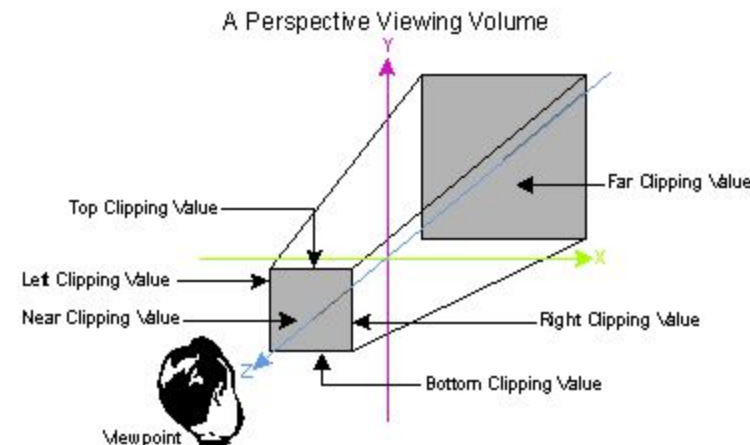
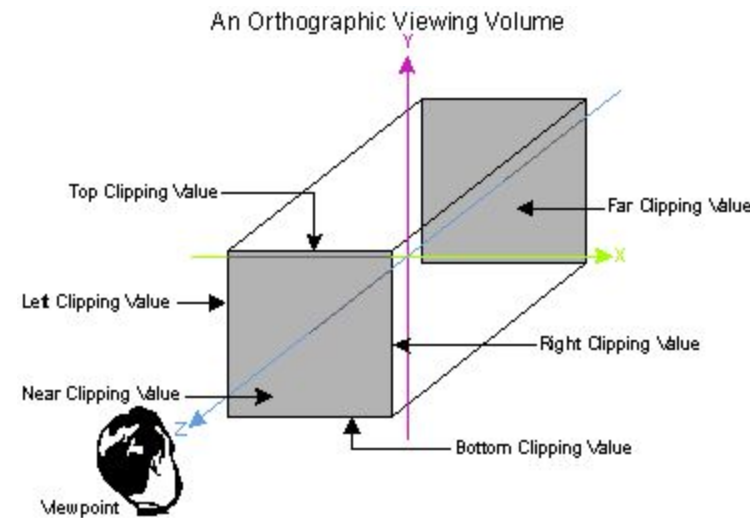


3D to 2D projection

3D projection

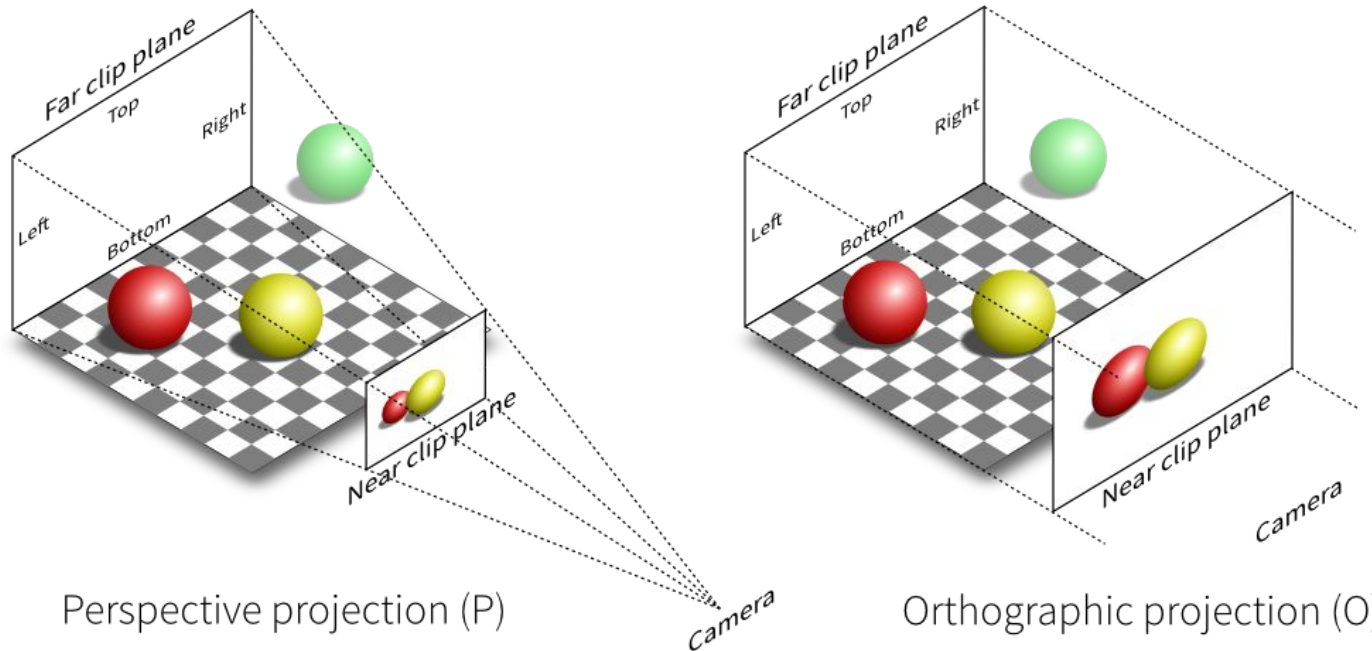
3D projection is any method of mapping three-dimensional points to a two-dimensional plane.

- Orthographic projection
 - ...plan,
 - bird's-eye,...
- Perspective projection



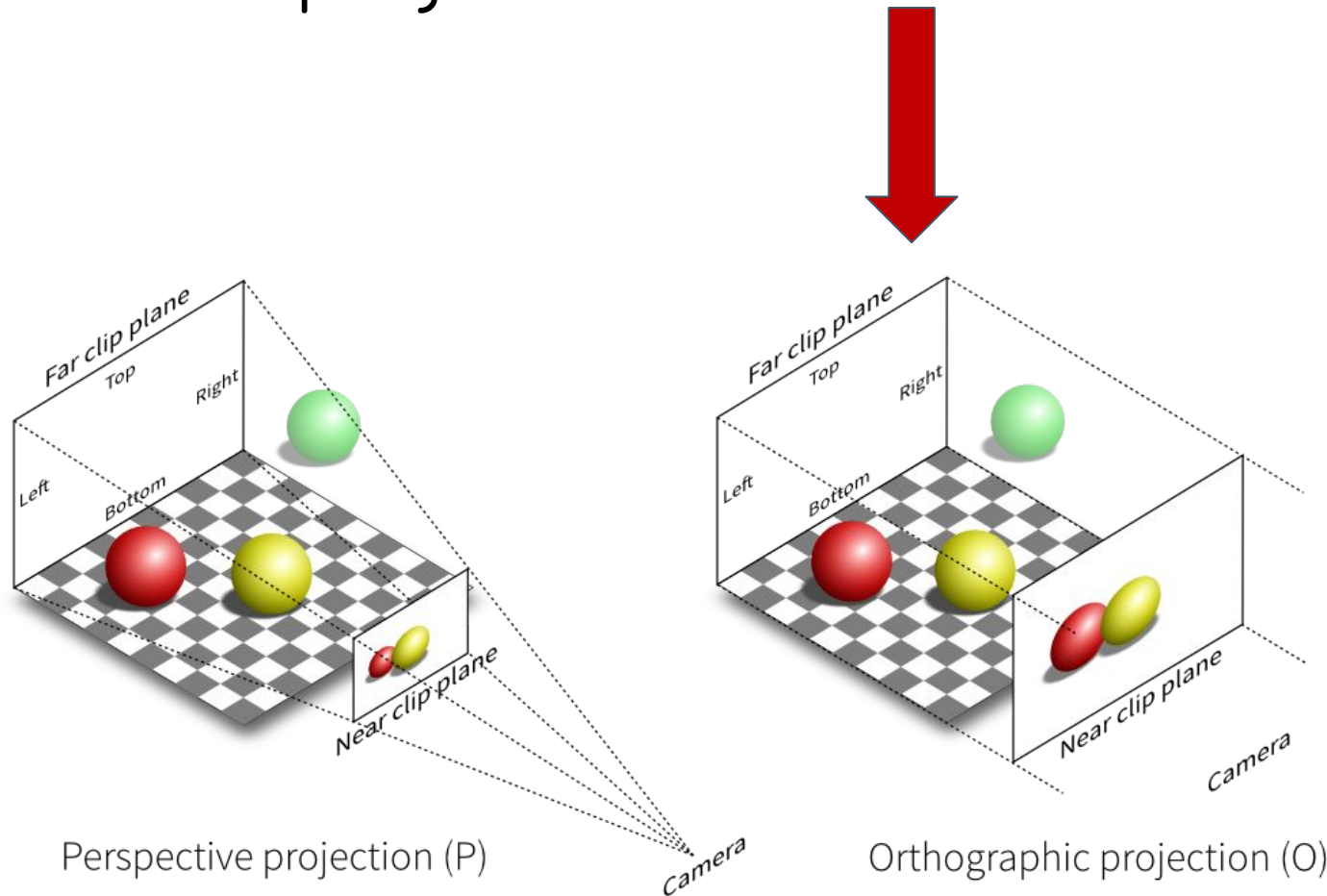
Perspective Projection

Parallel lines converge towards a vanishing point (úbežný bod -úbežník) on the horizon.



Orthographic Projection

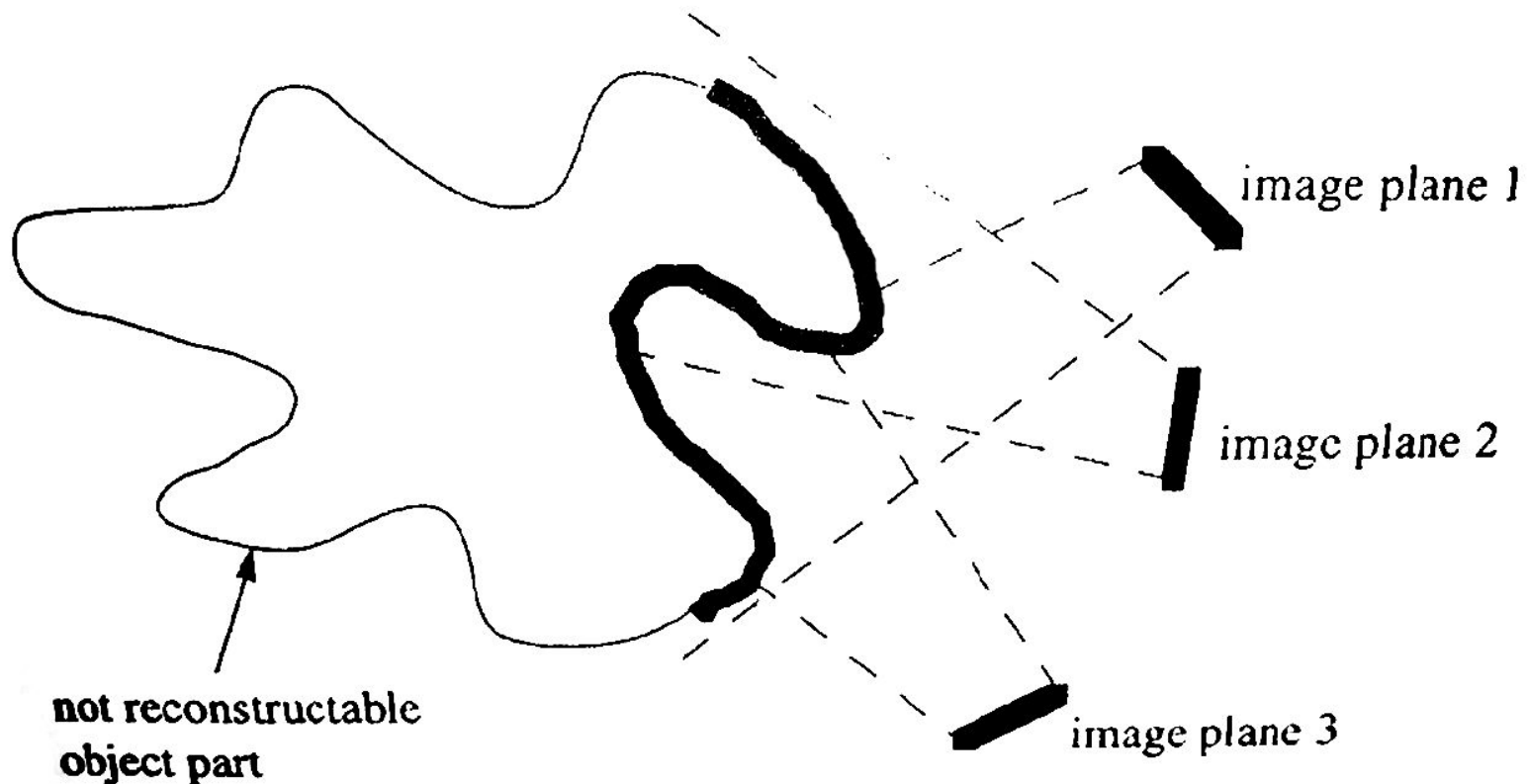
Lines that are parallel in 3D remain parallel in the 2D projection.



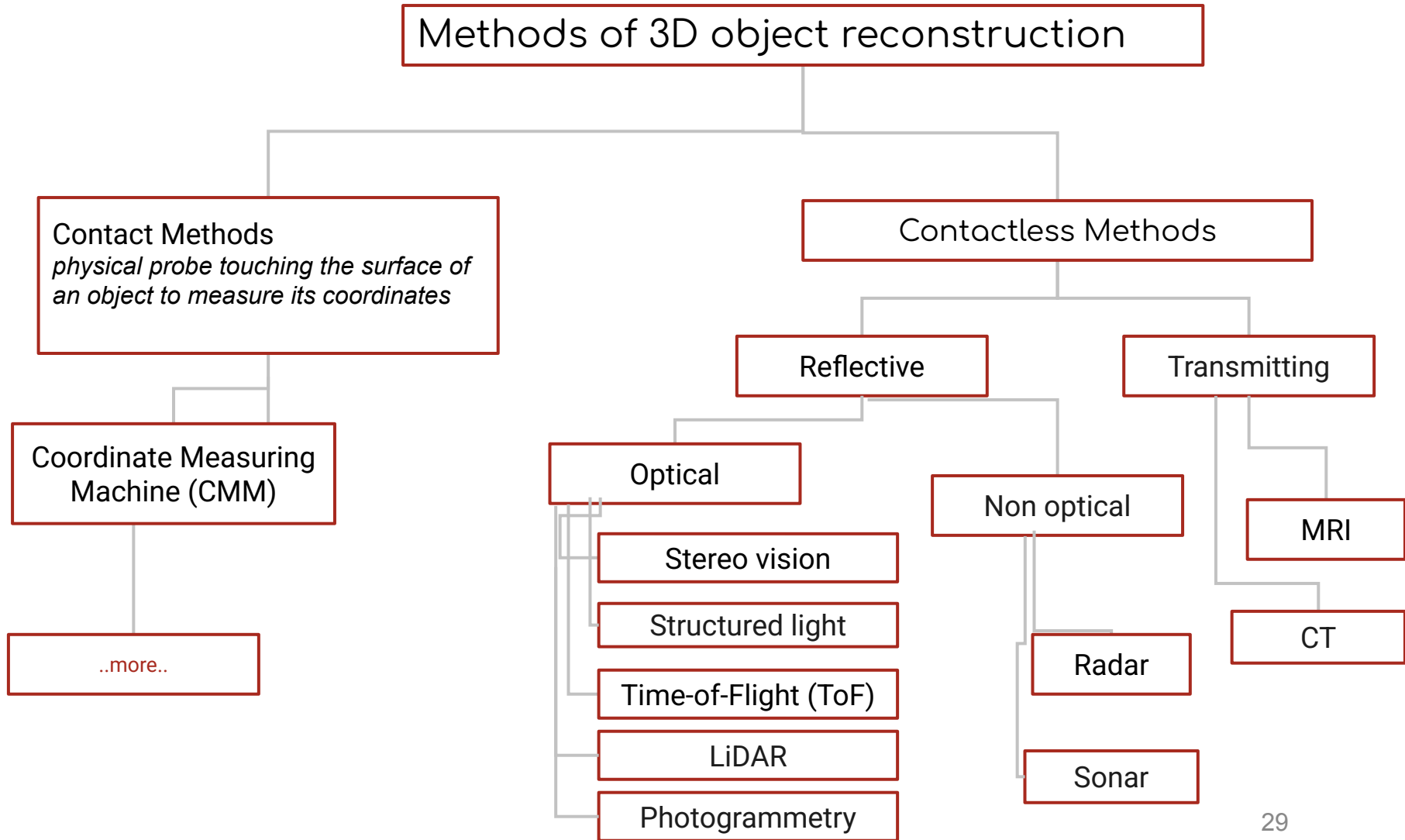
Methods of 3D reconstruction

2.5 D surface representation

Only projected object surfaces can be reconstructed –
restriction to 2.5 D representation



Methods of 3D reconstruction



Contact 3D scanner

Contact 3D scanners probe the subject through physical touch.

CMM (coordinate measuring machine) include three main components:

- The main structure which include three axes of motion
- Probing system (target for calibration)
- Data collection and Reduction system - typically includes a machine controller, desktop computer and application software.



3D Shape from ...

3D Shape from shading, from silhouette, form texture....

3D Shape from:

- shading
- silhouette
- texture

} used mainly in the research

-
- stereo
 - light striping
 - motion
 - TOF Time of Flight

} used in practice

Shape From Shading

first shape-from-shading (SFS) technique was developed by Horn in the early 1970s

pattern of shading
in a single image

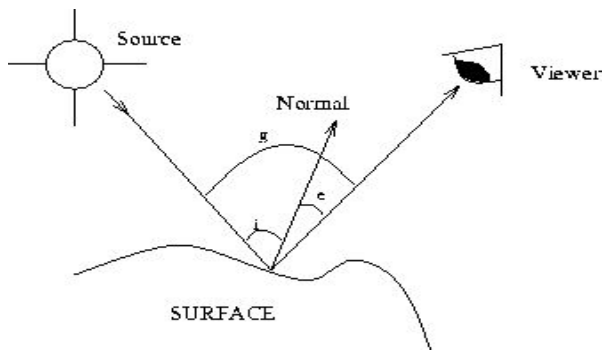
typical application
example are in astronomy



a) Image



b) 3D surface reconstructed
from the single image a)



Shape From Shading

Illumination: the irradiance and the direction of the illumination are known.

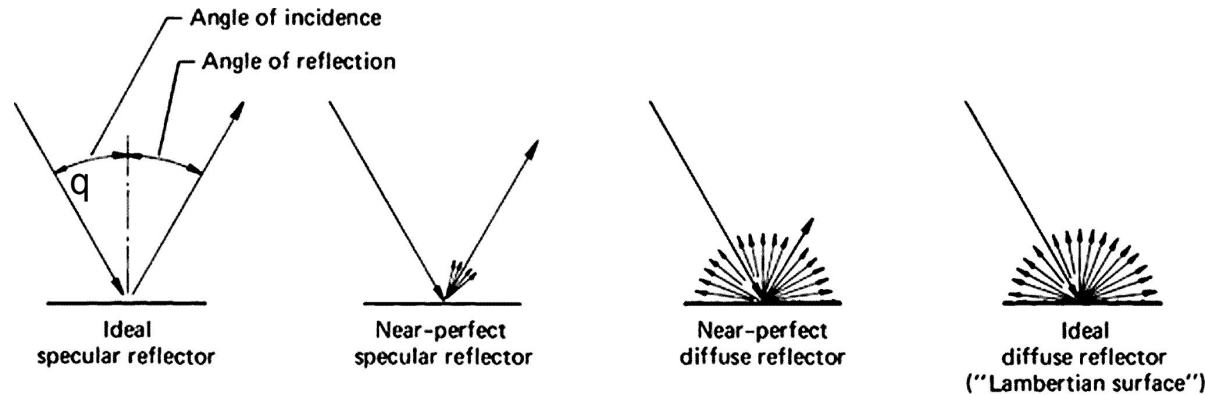
Reflection: the reflection properties of the object surface are known.

Illumination+reflection:

the interaction between a light source and reflection properties can be described by a **reflectance map** - > hence **a unique scene radiance value is assigned to each surface orientation**

Lambertian Reflectance Model

Difficulties with Shape from Shading



Lambertian Reflectance Model :

- Diffuse surfaces appear equally bright from all directions
- For point illumination, brightness proportional to $\cos(q)$

Difficulties with Shape from Shading :

- Robust estimation of lightness parameters
- Shadows
- Non-Lambertian surfaces
- More than 1 light, or "diffuse illumination"
- Interreflections

Deep architecture for Shape From Shading – example SfSNet

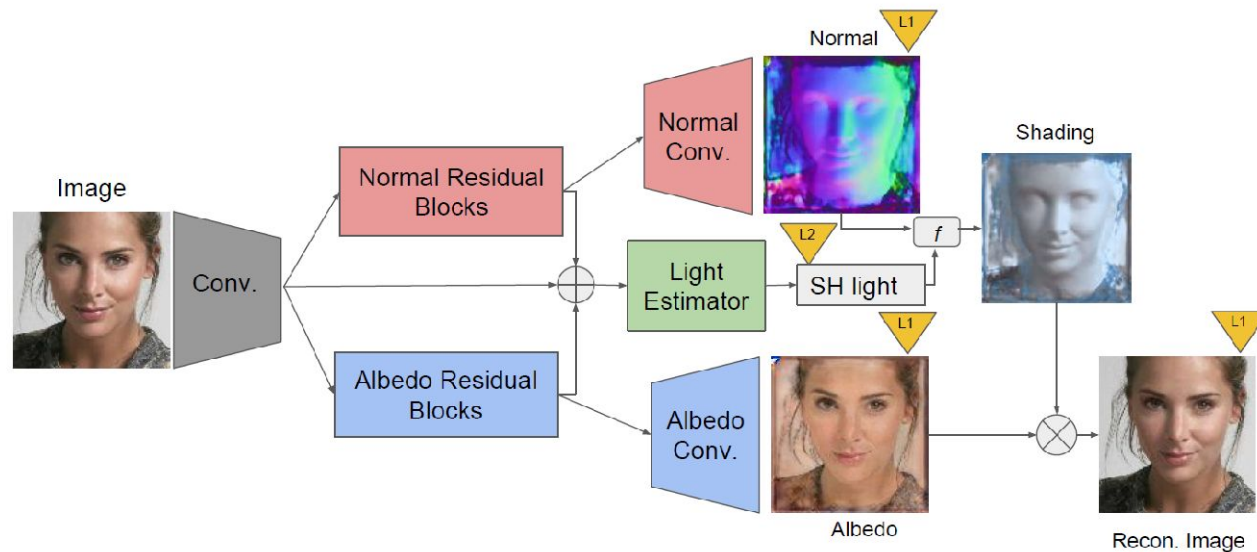


Figure : Network Architecture.

SfSNet consists of a novel decomposition architecture that uses residual blocks to produce normal and **albedo features**.

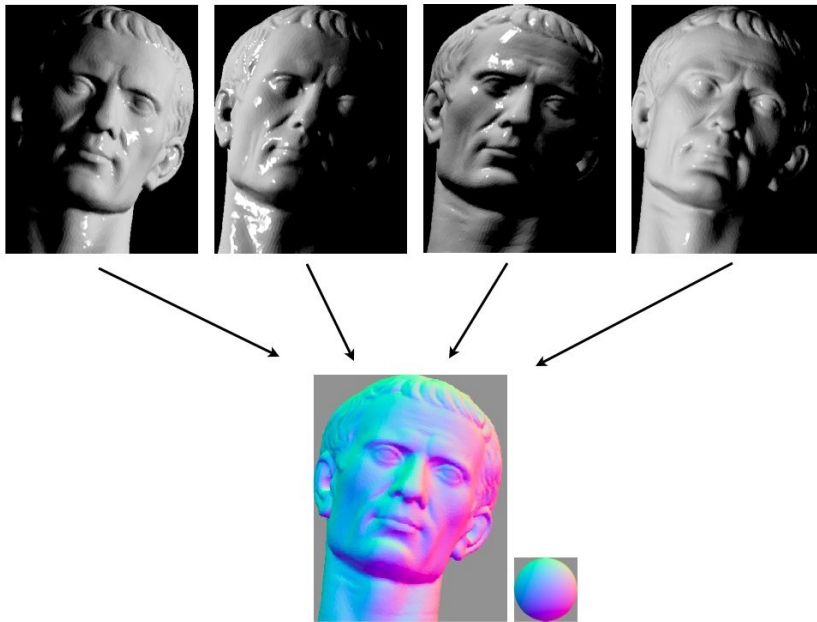
They are further utilized along with image features to estimate lighting, inspired by a physical rendering model. f combines normal and lighting to produce shading. (Best viewed in color)

...An albedo feature is a large area on the surface of a planet (or other solar system body) which shows a contrast in brightness or darkness (albedo) with adjacent areas.

Soumyadip Sengupta, Angjoo Kanazawa, Carlos D. Castillo, and David W. Jacobs University of Maryland, College Park, University of California, Berkeley, SfSNet: Learning Shape, Reflectance and Illuminance of Faces 'in the wild'

Photometric stereo method (PSM) – photometric stereo

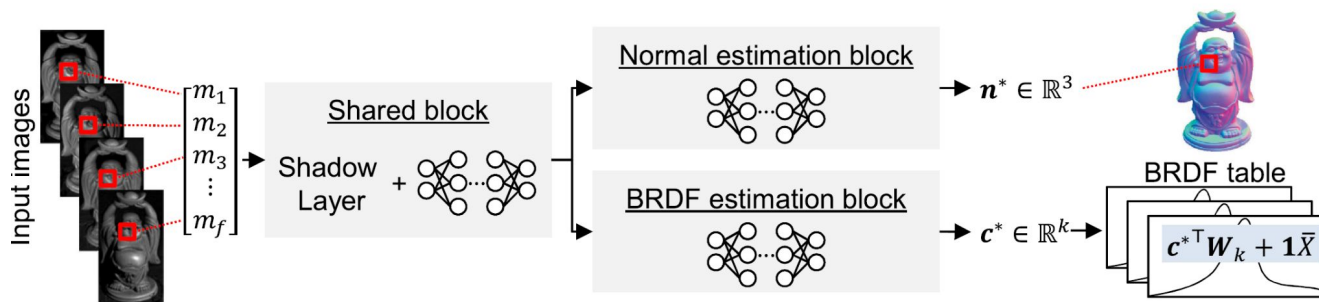
Photometric stereo is a technique for estimating the surface normals of objects by observing that object under different lighting conditions.



Photometric stereo method (PSM) – photometric stereo

- Extension of Shape from shading –
- Class of methods that use two or more images for shading based 3D shape reconstruction
- Object is consecutively illuminated by several light sources.

Deep photometric stereo network (example):



Overview of the example of a deep photometric stereo network (DPSN).

It consists of three components:

1. shared,
2. normal estimation,
3. BRDF estimation blocks.

The shared block is formed by a shadow layer and dense layers.

BRDF: bidirectional reflectance distribution functions

Shape From Texture

THE RECOGNITION MODEL IS AFFECTED BY THE WAY FROM THE WAKE-UP IN (the maximum likelihood estimate). It seems most likely therefore as with discrepancies result from a local "basin of attraction" is a learning that does not lead to a local maximum of the likelihood. Discrepancies can be eliminated for the model with a correlation correction in either of two ways. One way is to reduce the learning rate generative parameters (in the wake phase), while leaving the learning rate unchanged. As discussed in section 32, there is a natural reason to think that this method will lead to the maximum estimates, since the recognition model will then have time to learn the generative model perfectly. When the generative learning is reduced by setting $\eta = 0.00005$ and $\alpha = 0.99995$, the maximum estimates are indeed found in eight of eight test runs. The second is to impose a constraint that prevents the generative parameters

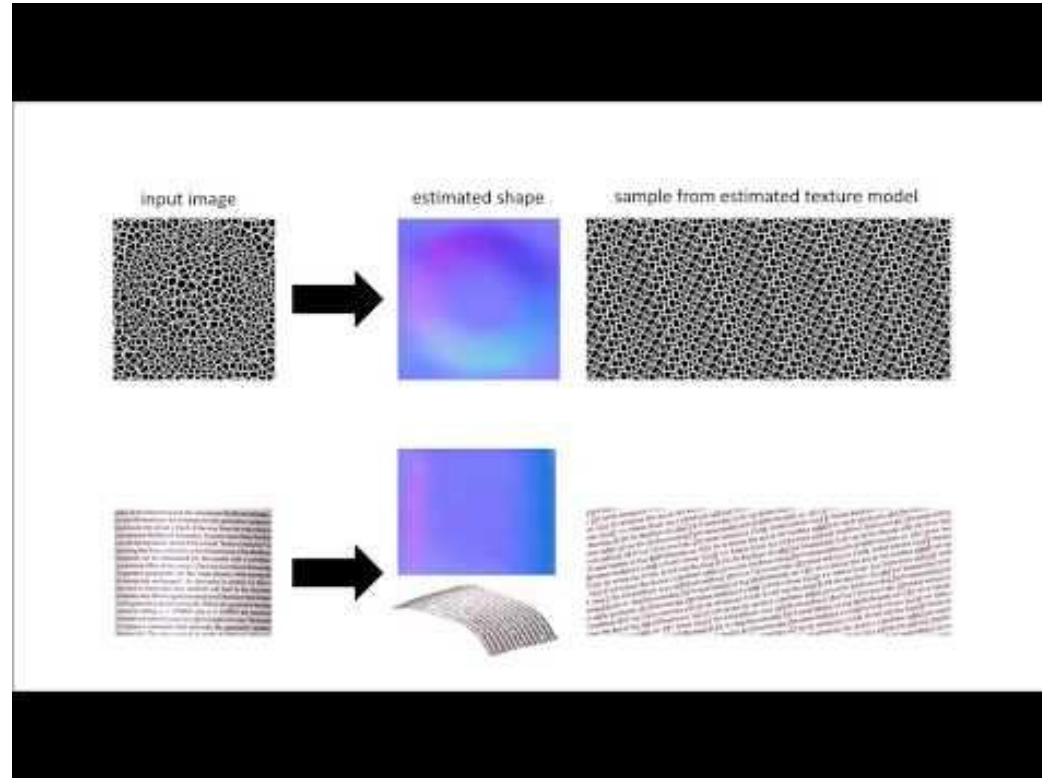


THE RECOGNITION MODEL IS AFFECTED BY THE WAY FROM THE WAKE-UP IN (the maximum likelihood estimate). It seems most likely therefore as with discrepancies result from a local "basin of attraction" is a learning that does not lead to a local maximum of the likelihood. Discrepancies can be eliminated for the model with a correlation correction in either of two ways. One way is to reduce the learning rate generative parameters (in the wake phase), while leaving the learning rate unchanged. As discussed in section 32, there is a natural reason to think that this method will lead to the maximum estimates, since the recognition model will then have time to learn the generative model perfectly. When the generative learning is reduced by setting $\eta = 0.00005$ and $\alpha = 0.99995$, the maximum estimates are indeed found in eight of eight test runs. The second is to impose a constraint that prevents the generative parameters



Shape and texture from a single image (left). The output is a 2.5D shape (two middle columns) and a flat-texture generative process (right). The approach succeeds for a wide variety of textures, including those for which previous methods break down.

Example:
<http://vision.seas.harvard.edu/sft/>



Shape From Focus

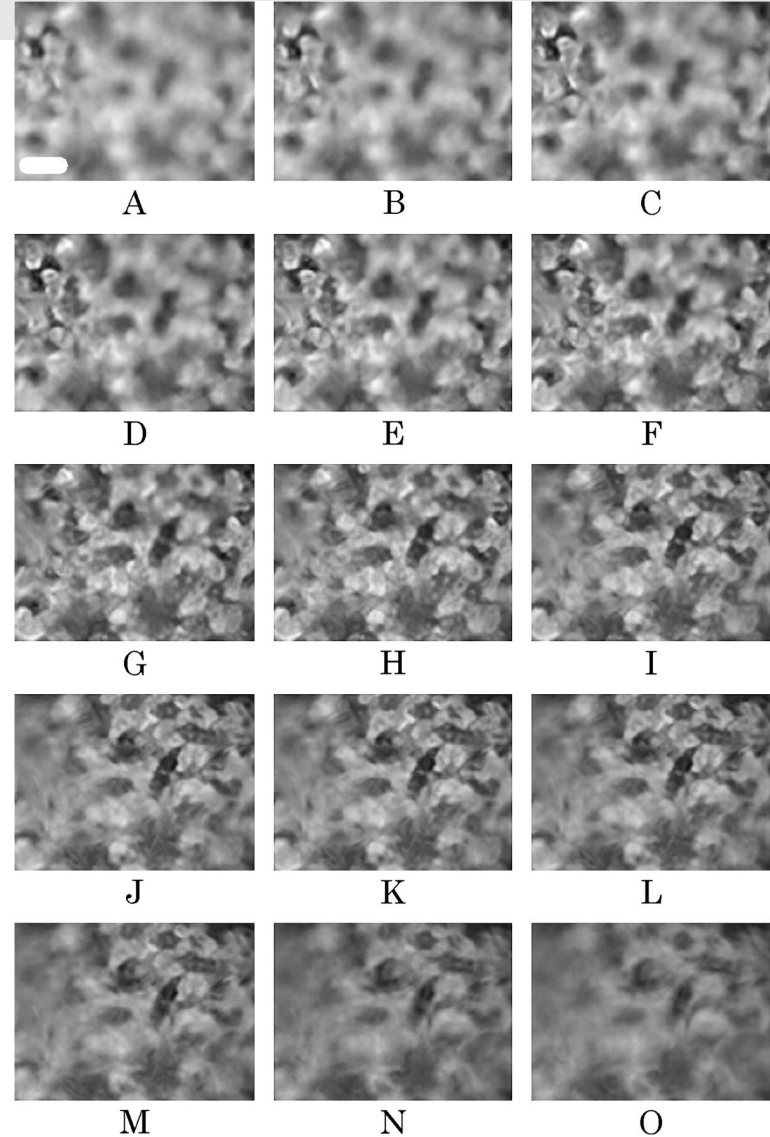
Depth from blur

-derived by different camera settings (focus)

Typical application-

Micrographs (microscopical images)

Example : A series of unprocessed chlorophyll fluorescence (F) images



Structured lighting

Structured lighting

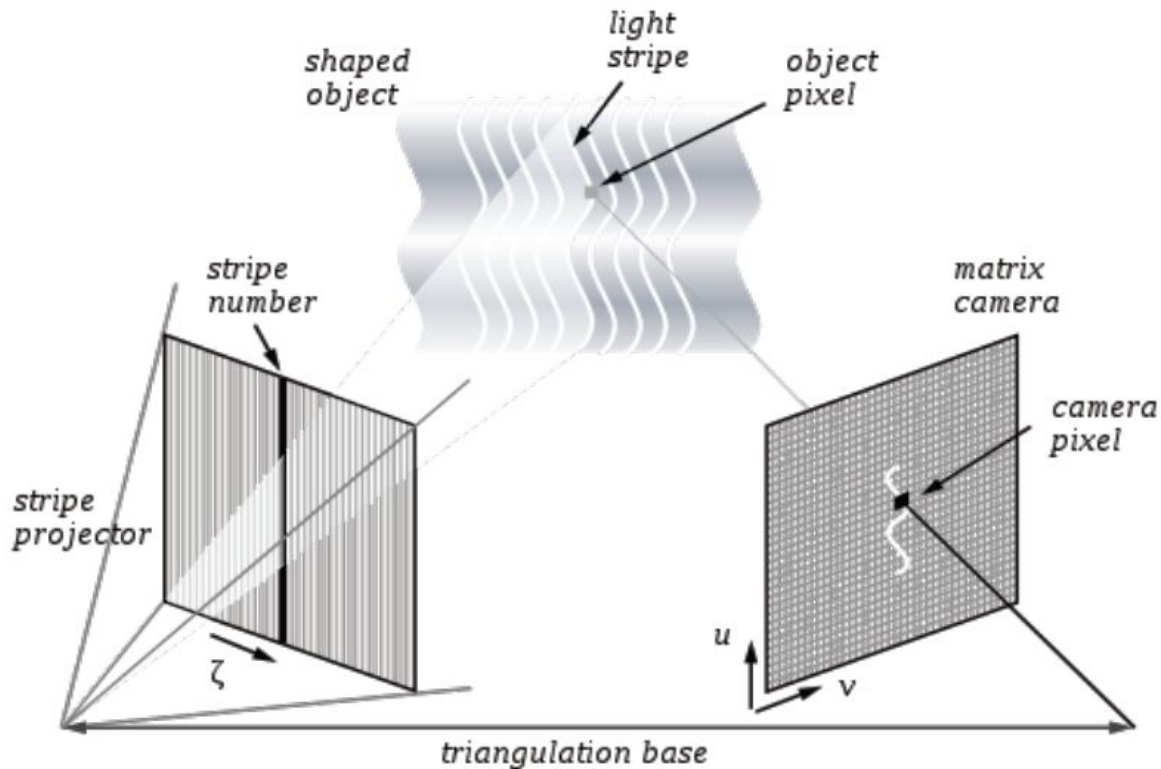
Laser scanner



Structured lighting

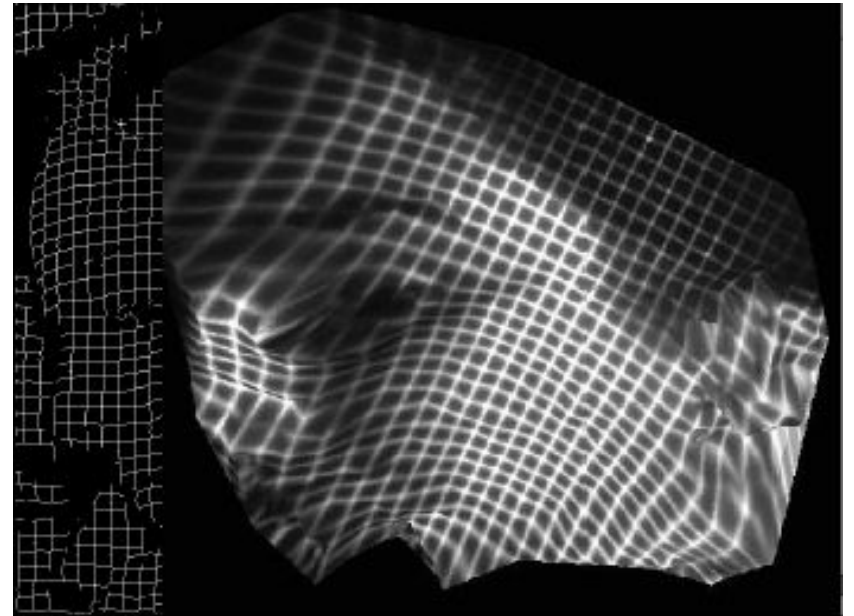
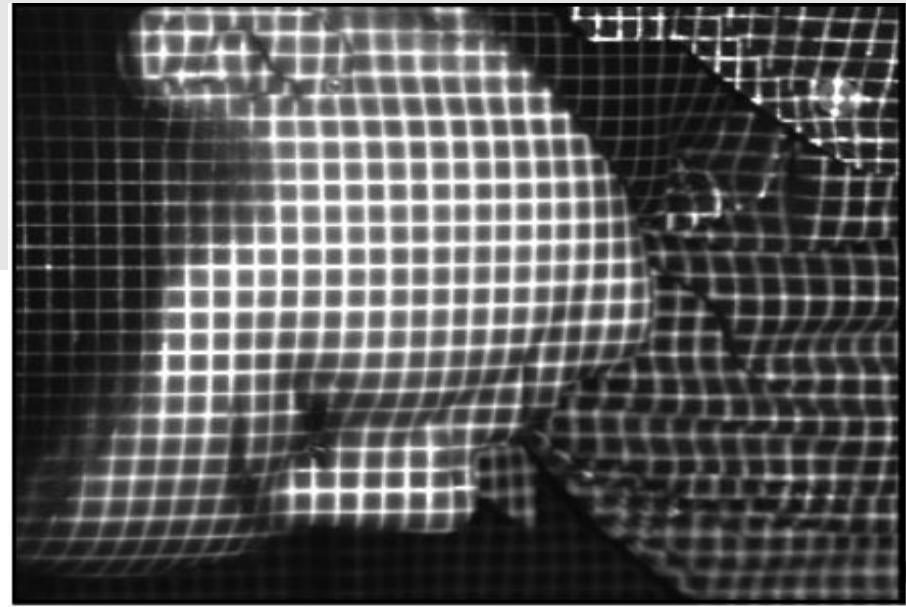
Projection of stripe pattern

Displacement $\rightarrow$ 3D information



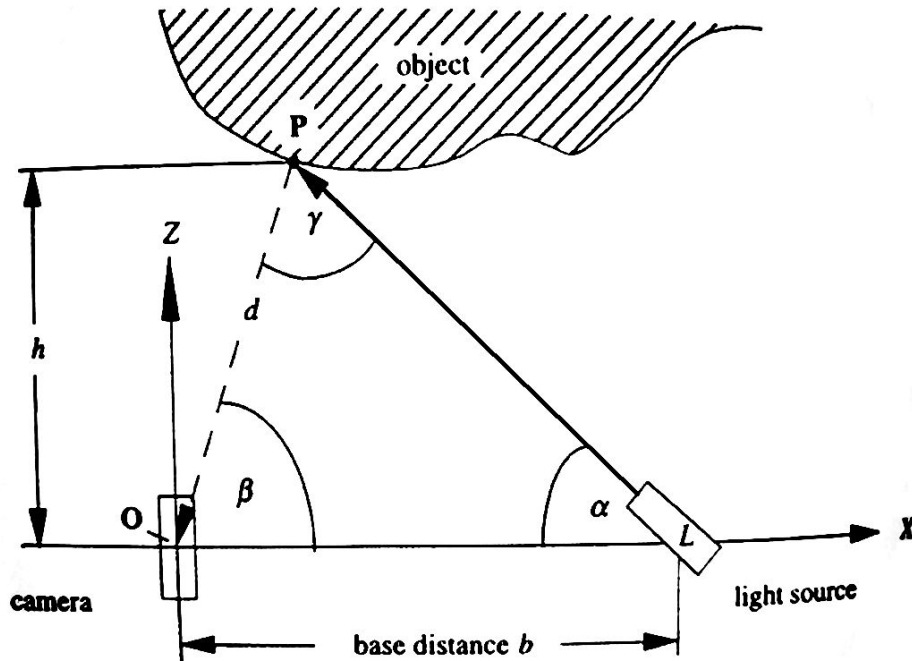
Structured Lighting

A structured-light 3D scanner is a device for measuring the three-dimensional shape of an object using projected light patterns and a camera system



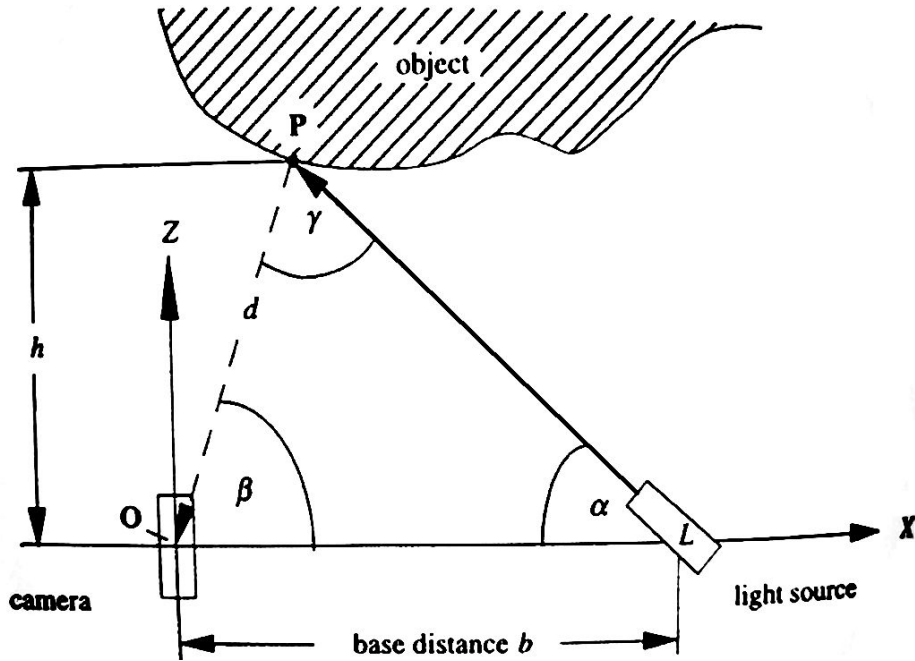
Light spot projection

Simple method to measure distances between the image acquisition system and points on the object in the scene.

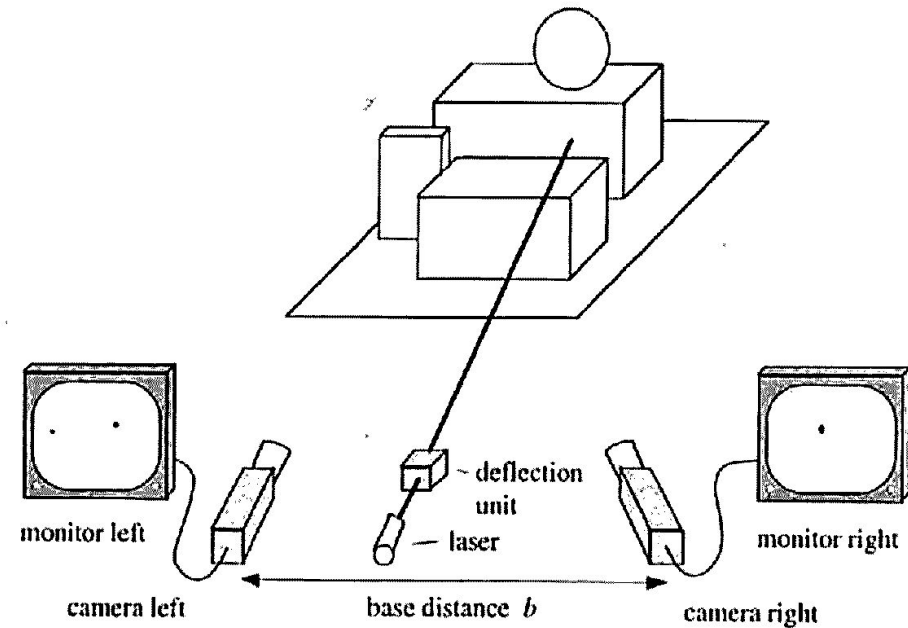


Triangulation with a light spot projector

Light spot analysis / Light spot stereo analysis



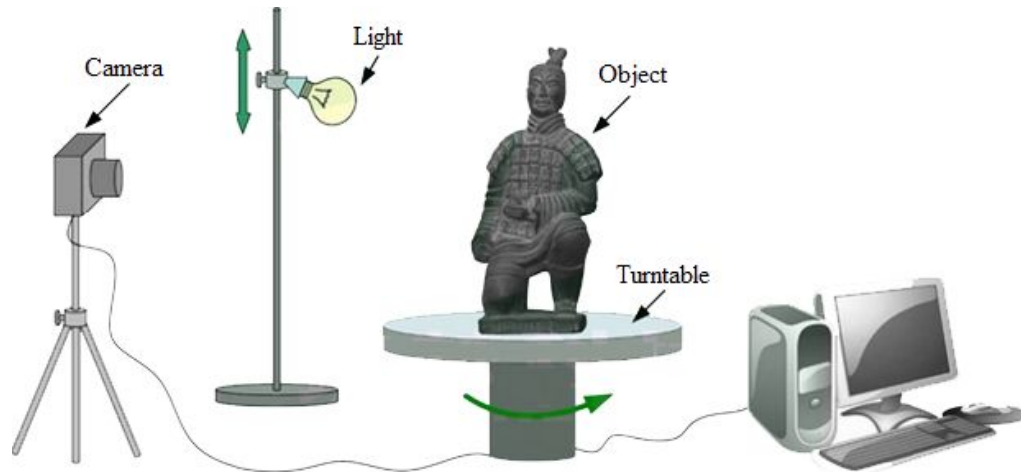
Arrangement for a 3D method based on light spot analysis



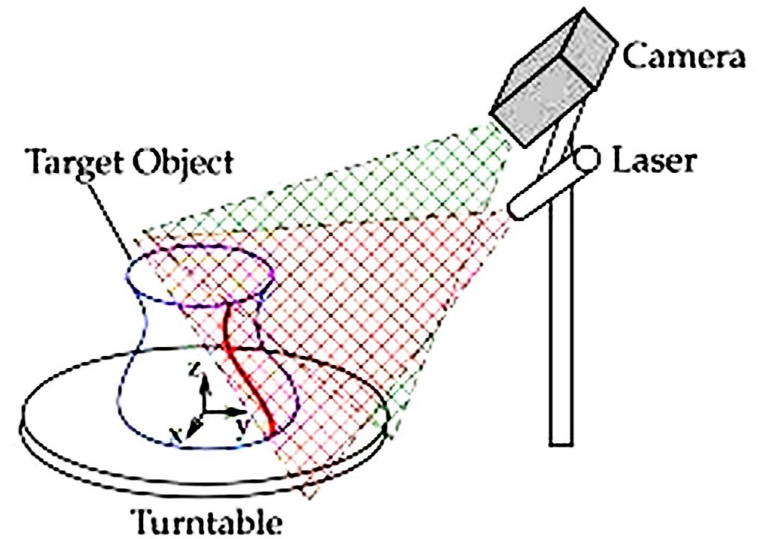
Arrangement for a 3D method based on light spot stereo analysis

Example - turntable data acquisition

Laboratory set-up for shape reconstruction



Using light source



Using laser point

Light Coding

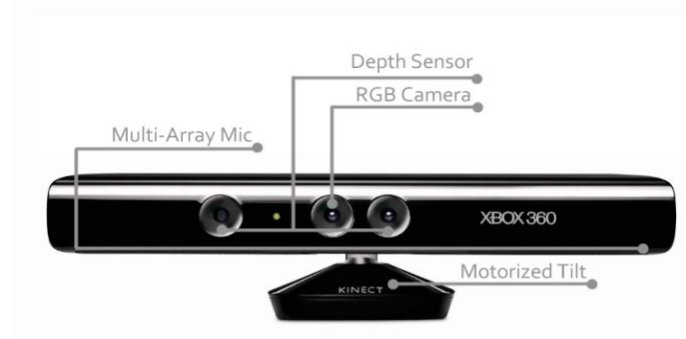
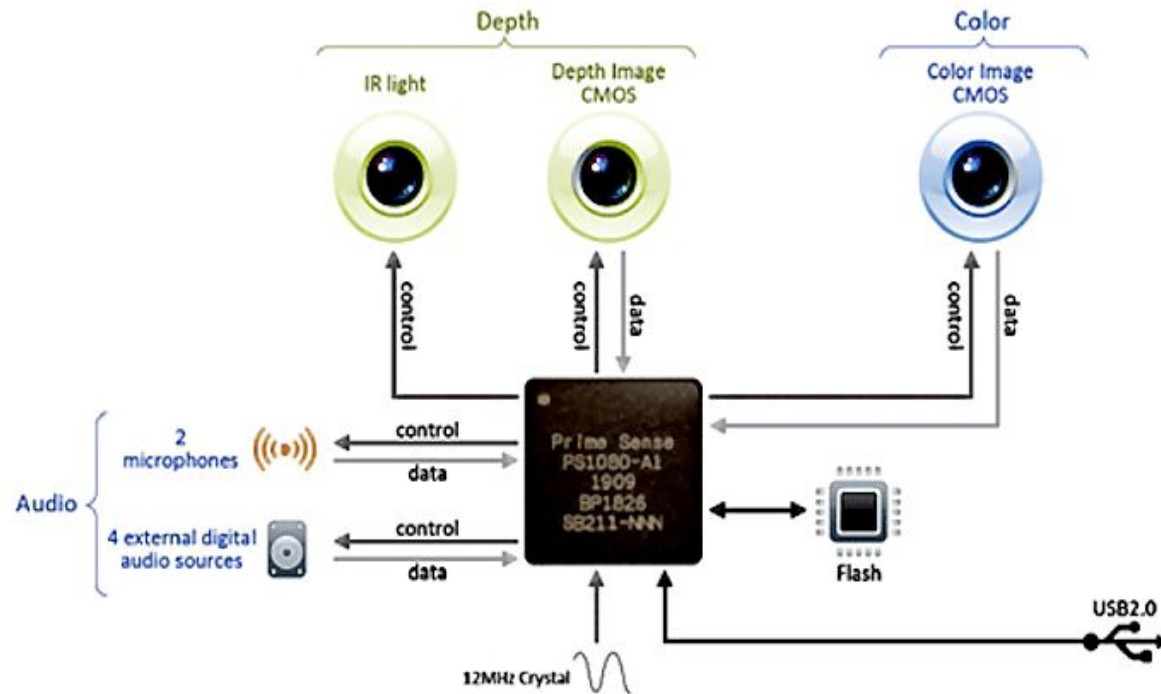
Light Coding

Projection of light pattern

Comparing of pattern area with calibrated pattern

Example - Kinect

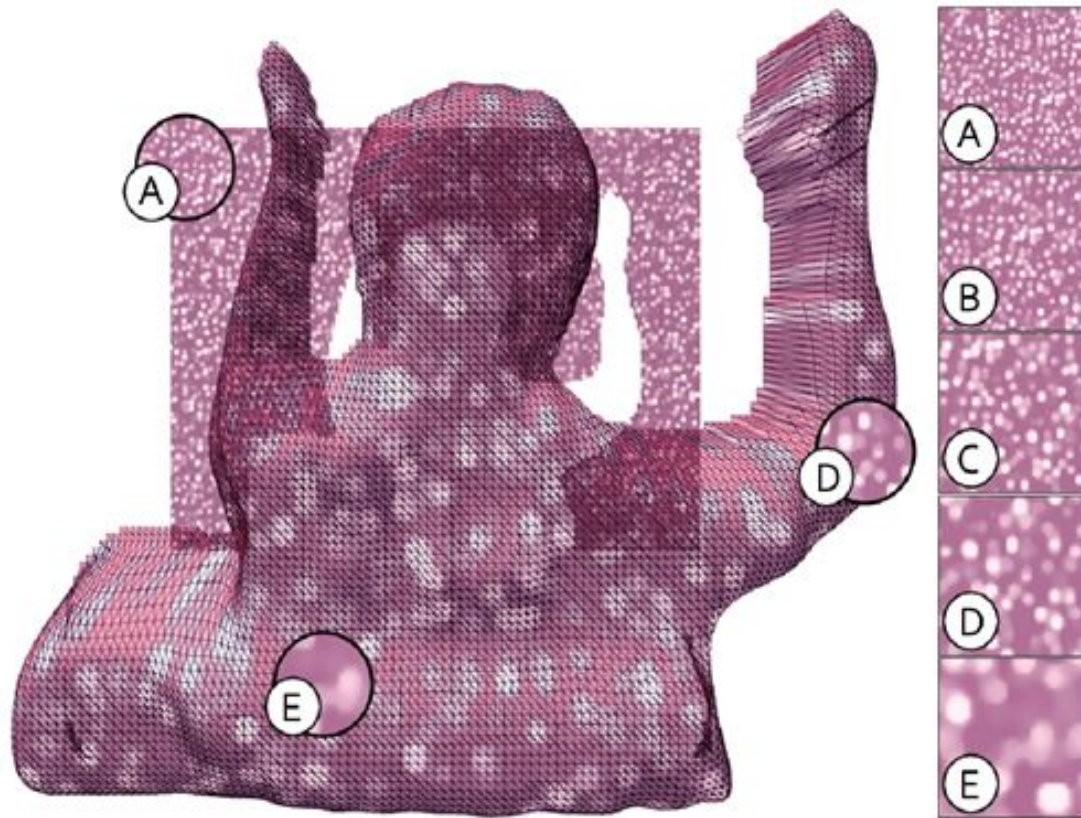
Kinect



Zdroj: Kinect_tech

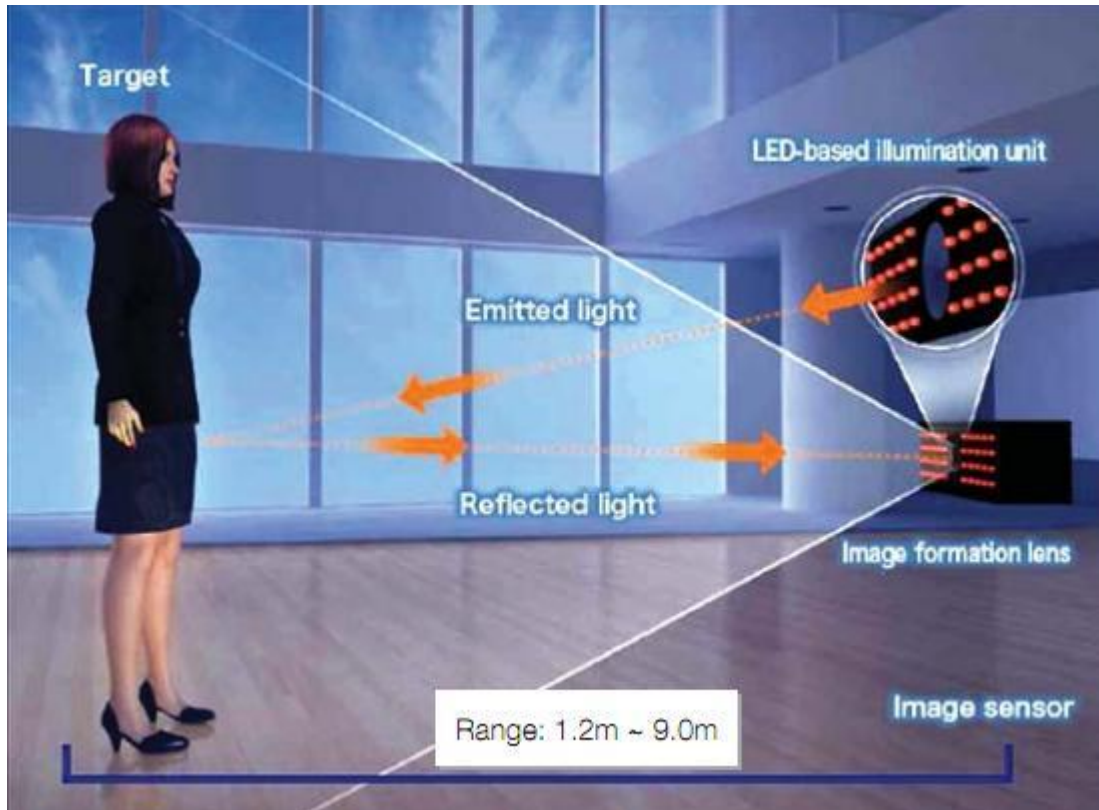
CADET Center for Advances in Digital Entertainment Technologies

Kinect - Infrared spots size mapping



Time-of-flight (TOF)

Time-of-flight (TOF)

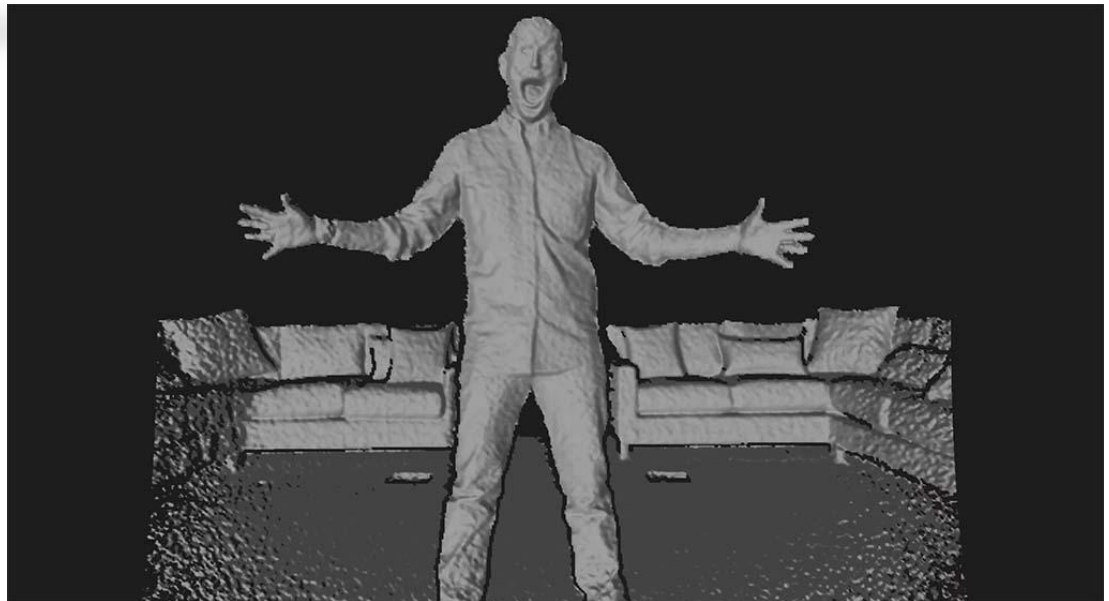


Time-of-flight (TOF) camera

Infrared light from the camera's internal lighting source is reflected by objects in the scene and travels back to the camera, where its precise time of arrival is measured independently by each of tens of thousands of sensor pixels.



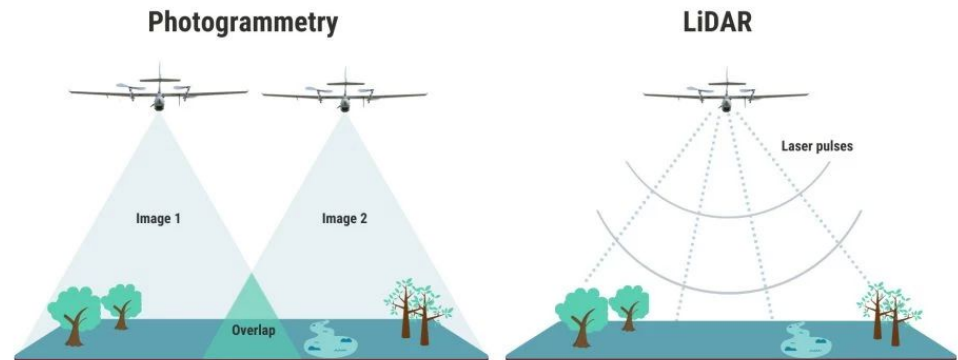
KinectV2



LiDAR (Light detection and ranging)

Lidar — Light Detection and Ranging — is a remote sensing method used to examine the surface of the Earth.

- primarily uses a laser sensor (or distance sensor).
- A lidar instrument principally consists of a laser, a scanner, and a specialized GPS receiver.



LiDAR (Light detection and ranging)

Compared to cameras, ToF, and other sensors, lasers are significantly more precise, and are used for applications with high-speed moving vehicles such as self-driving cars and drones.

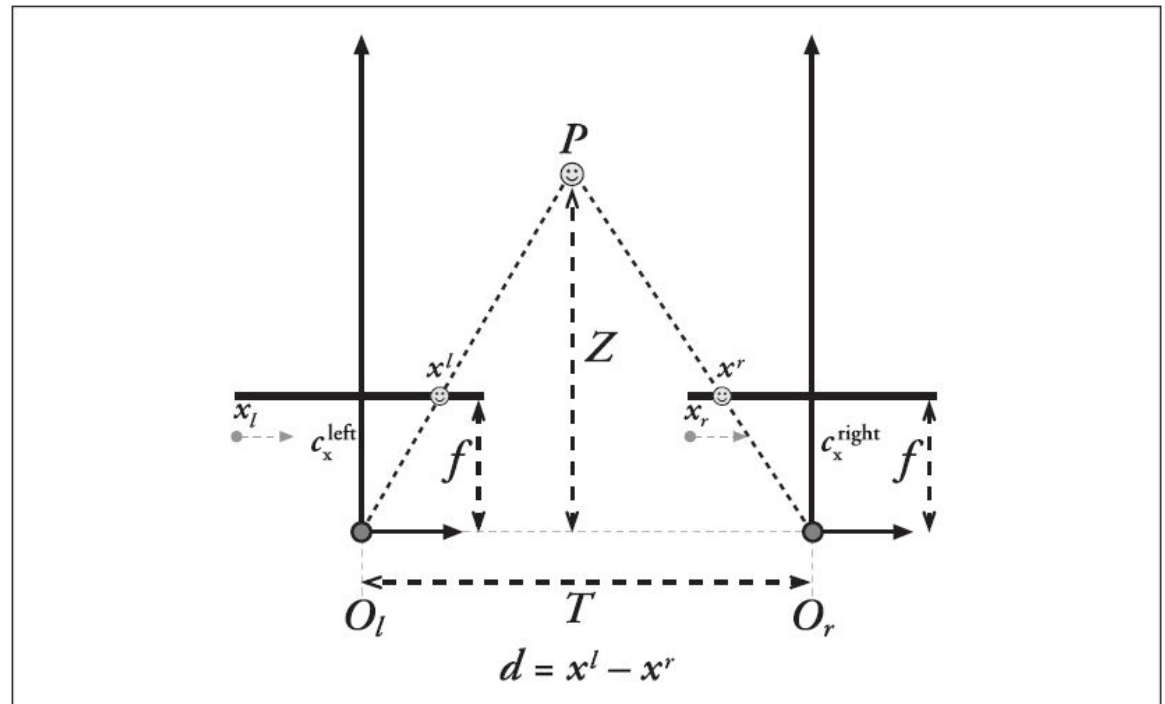
The output values from laser sensors are generally 2.5D (x, y, d) or 3D (x, y, z) point cloud data.

Stereo acquisition

Stereo - a simplified case

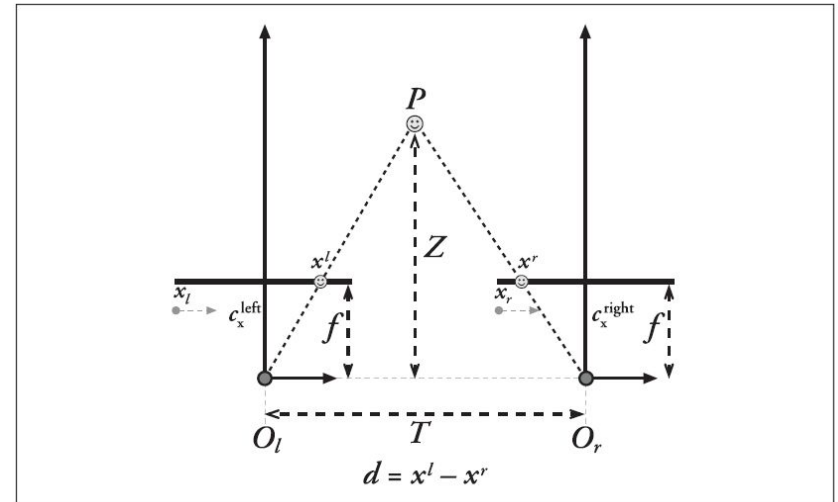
If we have a perfectly undistorted and aligned stereo cameras and known correspondence $\rightarrow$ the depth Z can be found by similar triangles;

the principal rays of the imagers begin at the centers of projection



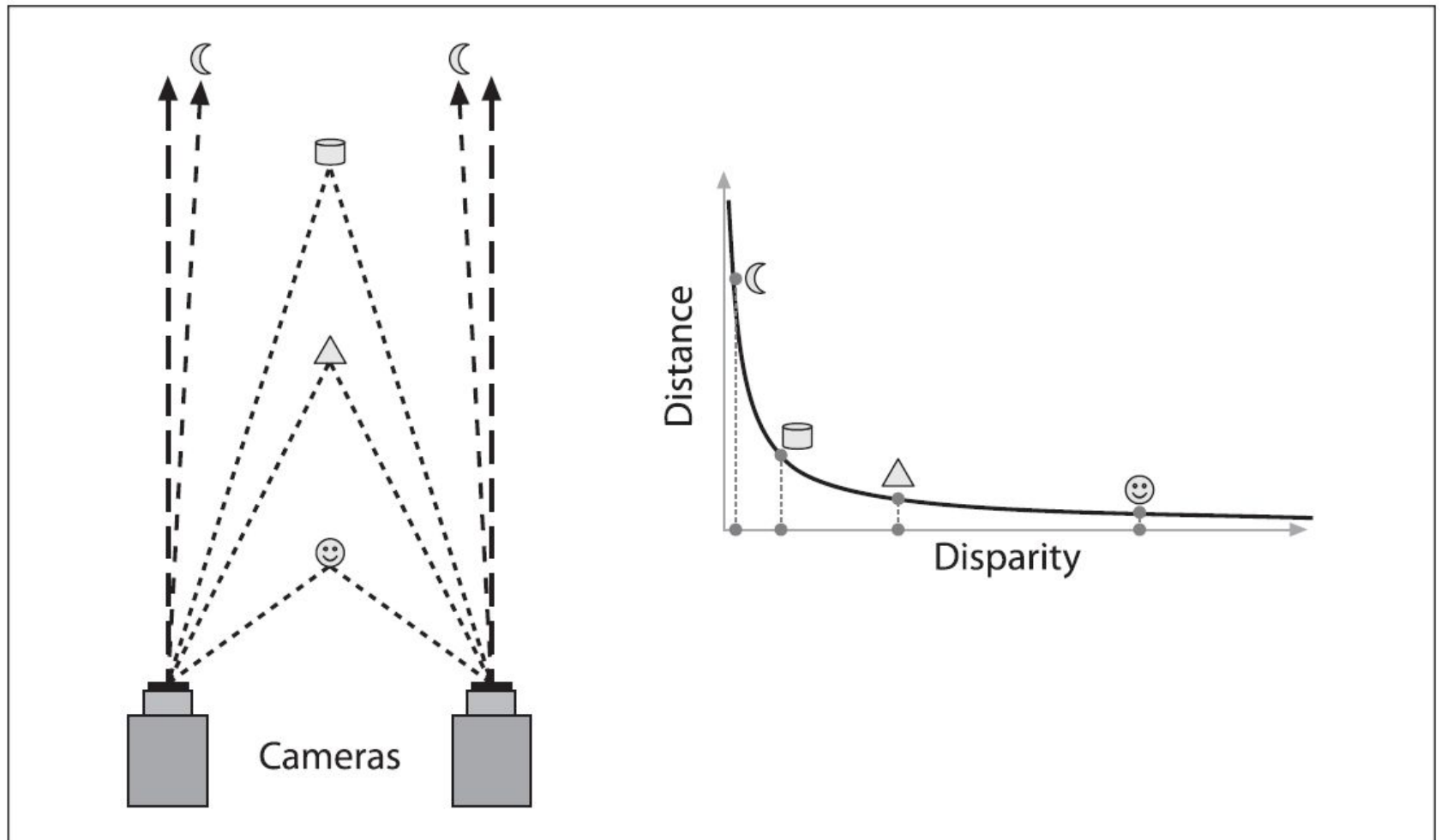
Disparity versus distance

the depth is **inversely proportional** to the disparity between these views, where the disparity is defined simply by $d = x_l - x_r$



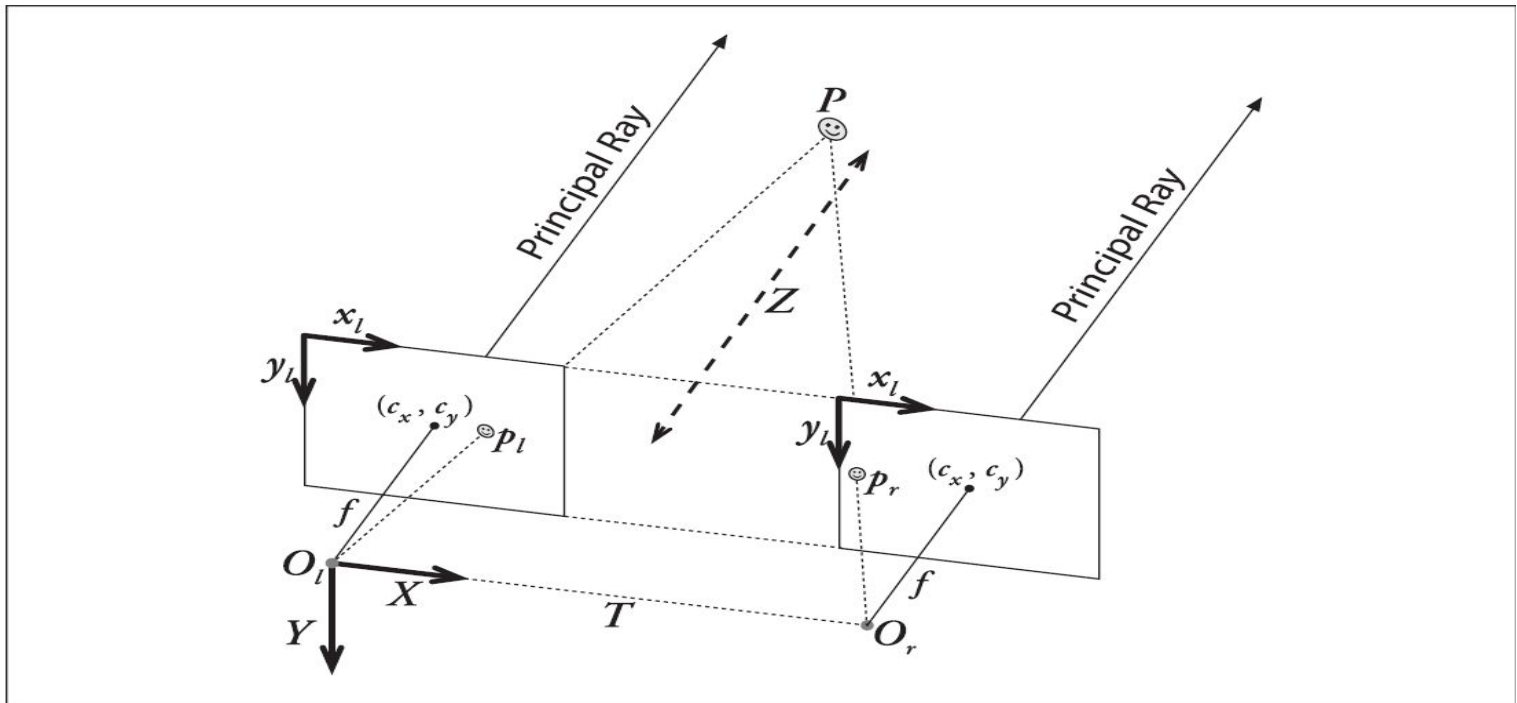
$$\frac{T - (x^l - x^r)}{Z - f} = \frac{T}{Z} \quad \Rightarrow \quad Z = \frac{fT}{x^l - x^r}$$

Disparity versus distance



Stereo imaging

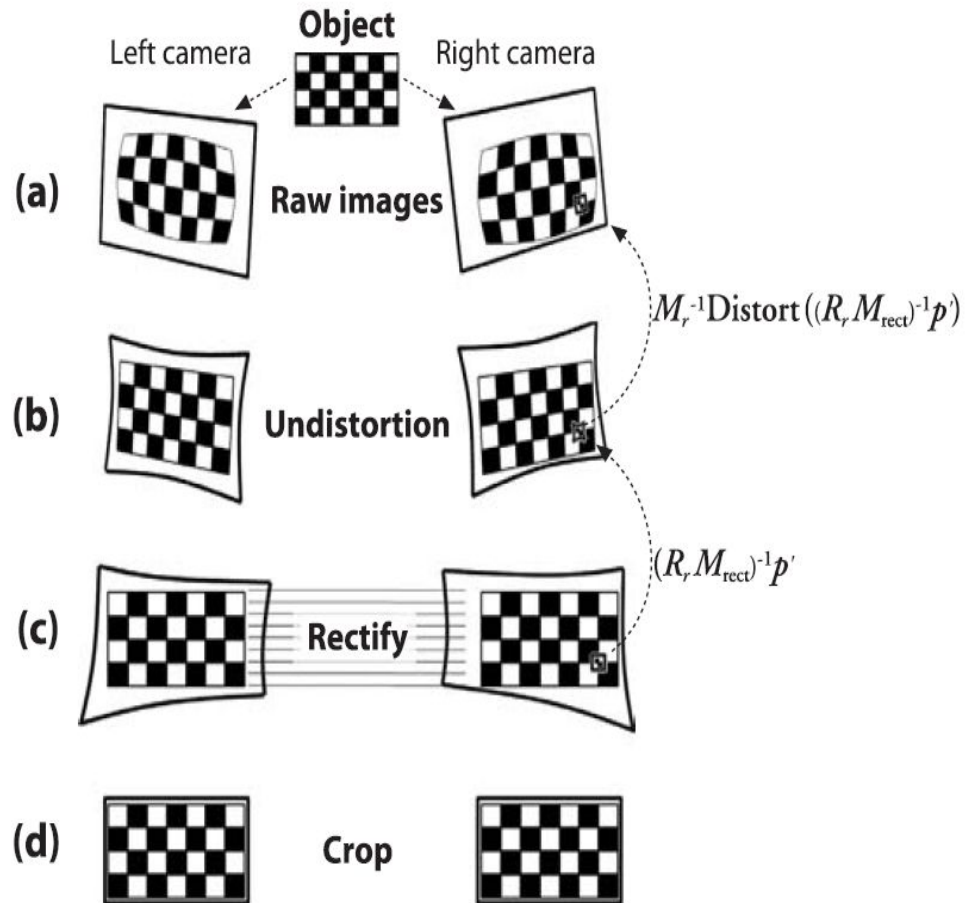
Stereo coordinate system used for undistorted rectified cameras



Stereo rectification

Stereo rectification: for

- the raw image (a)
 - is undistorted (b)
 - rectified (c)
 - finally cropped (d)
- to focus on overlapping cameras



Stereo imaging using two cameras : 4 steps (1-2)

1. Undistortion

Mathematically remove radial and tangential lens distortion. The outputs of this step are **undistorted images**.

2. Adjust for the angles and distances between cameras, a process called rectification.

The outputs of this step are images that are **row-aligned and rectified**.

Stereo imaging

using two cameras: 4 steps (3-4)

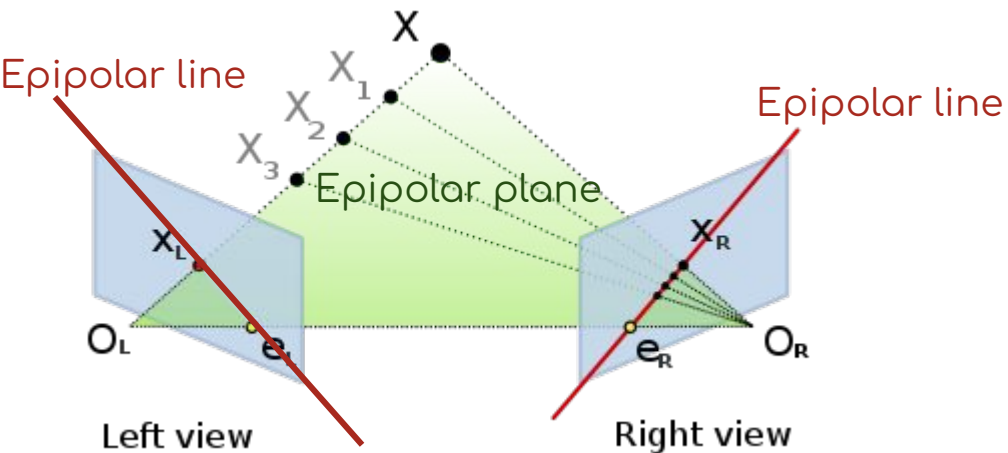
3. Find the same features (using local features or other method...) in the left and right camera views, a process known as correspondence. The output of this step is a **disparity map**, where the disparities are the differences in x-coordinates on the image planes of the same feature viewed in the left and right cameras: $x_l - x_r$.

4. If we know the geometric arrangement of the cameras, then we can turn the **disparity map into distances by triangulation**. This step is called **reprojection**, and the output is a **depth map**.

The geometry of stereo imaging

Epipolar geometry

- Projective geometry between two views of stereo vision.
- Independent of scene structure
- Depends only on the cameras' internal parameters and relative pose of cameras



The projection matrix

The projection matrix is used to convert from 3D real world coordinates to 2D image coordinates

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D Image Coordinates Intrinsic properties (Optical Centre, scaling) Extrinsic properties (Camera Rotation and translation) 3D World Coordinates

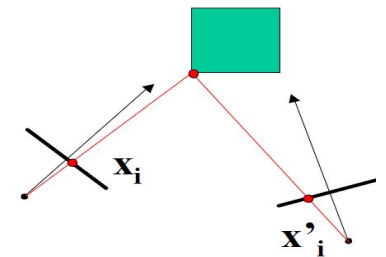
Epipolar Geometry and the Fundamental Matrix

- Projective geometry between two views
- Independent of scene structure
- Depends only on the cameras' internal parameters and relative pose of cameras
- Fundamental matrix F encapsulates this geometry

$$\mathbf{x}_i'^T \mathbf{F} \mathbf{x}_i = 0$$

for any pair of corresponding points $\mathbf{x}_i$ and $\mathbf{x}_i'$ in the 2 images

The fundamental matrix also tells how pixels (points) in each image are related to epipolar lines in the other image.



Fundamental Matrix F

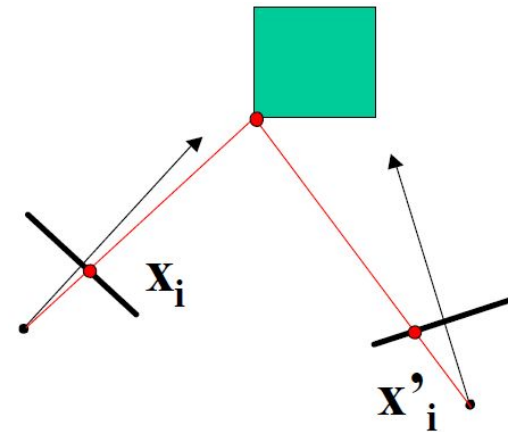
F can be computed from correspondences between image points alone.

F has 7 degrees of freedom

In principal F has 7 parameters up to scale and can be estimated from 7 point correspondences

$$\mathbf{x}_i'^T \mathbf{F} \mathbf{x}_i = 0$$

To use image coordinates we must consider the INTRINSIC camera parameters



References

• Multiple View Geometry in Computer Vision,
R. Hartley and A. Zisserman, Cambridge University Press, 2000.

The essential matrix E

The essential matrix is the specialization of the fundamental matrix to the case of **normalized image coordinates**

Fundamental matrix F_x and essential matrix E are 3×3 matrices which relate corresponding points in stereo images.

The fundamental matrix F has 7 degrees of freedom.
The essential matrix E has 5 degrees of freedom
fewer degrees of freedom (compared with F)

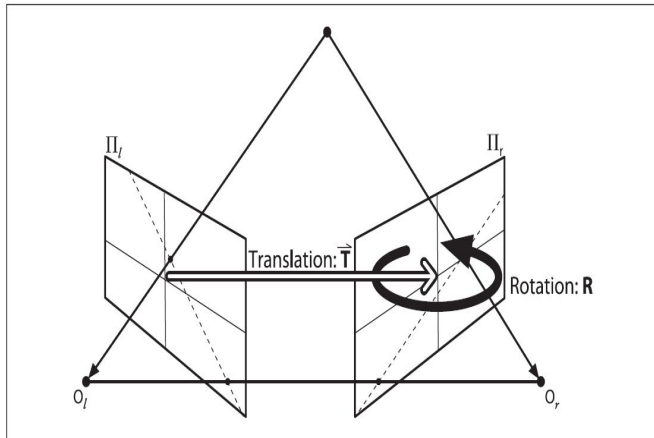
References

• Multiple View Geometry in Computer Vision,
R. Hartley and A. Zisserman, Cambridge
University Press, 2000.

Essential matrix E

The essential geometry of stereo imaging is captured by the **Essential matrix E** (3×3 matrix), which contains all of the information about the translation **T** and the **rotation R** .

Essential matrix can only be used in relation to **calibrated cameras**!



Essential matrix E

The Essential matrix can be seen as a precursor to the Fundamental matrix.

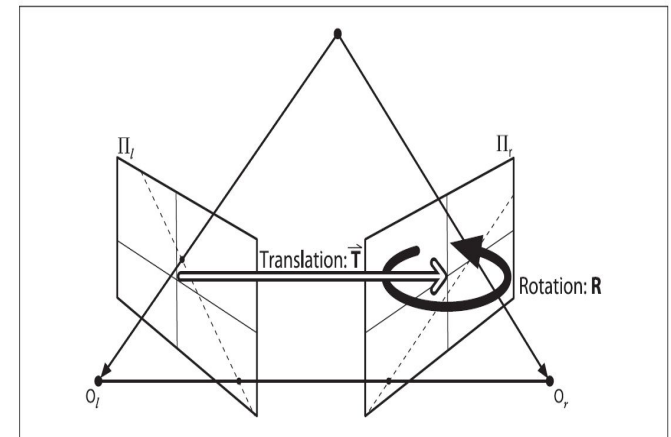
Both matrices can be used for establishing constraints between matching image points, but the essential matrix can only be used in relation to calibrated cameras since the inner camera parameters must be known in order to achieve the normalization.

If, however, the cameras are calibrated the essential matrix can be useful for determining both:

- the relative position and orientation between the cameras

and

- the 3D position of corresponding image points.

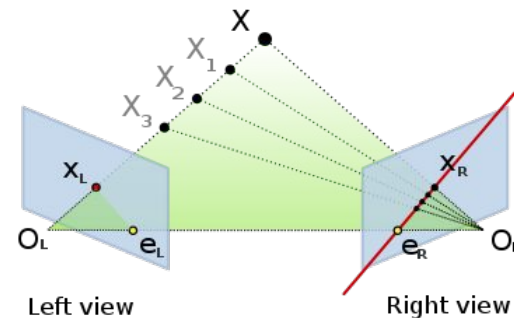


Essential and fundamental matrices

The essential and fundamental matrices are 3×3 matrices that “encode” the epipolar geometry of two views.

Motivation ->

Given a point in one image, multiplying by the essential/fundamental matrix will tell us which epipolar line to search along in the second view.

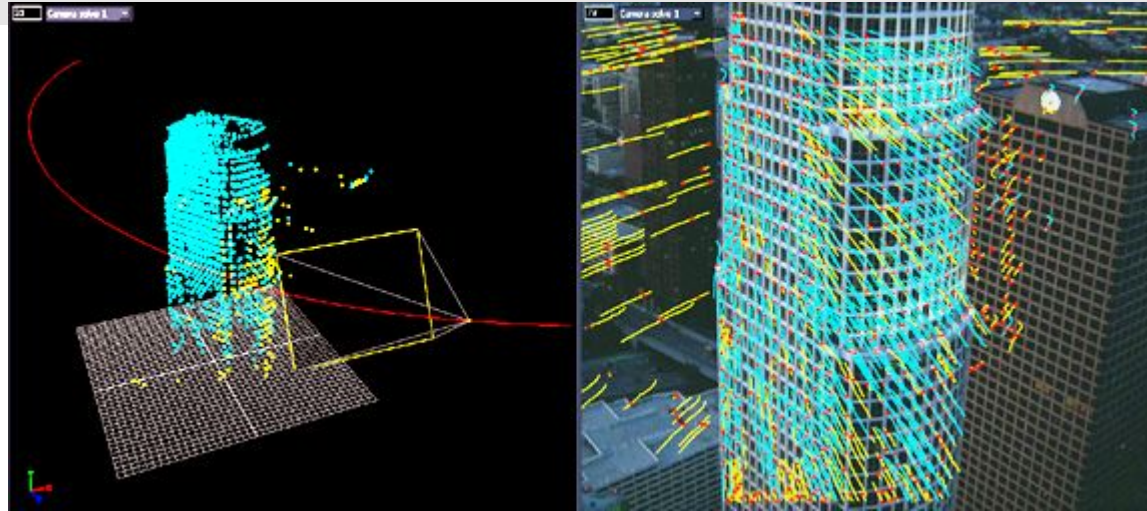


Structure From Motion SFM

Structure From Motion - SFM

Multiple Views

Video pipeline...

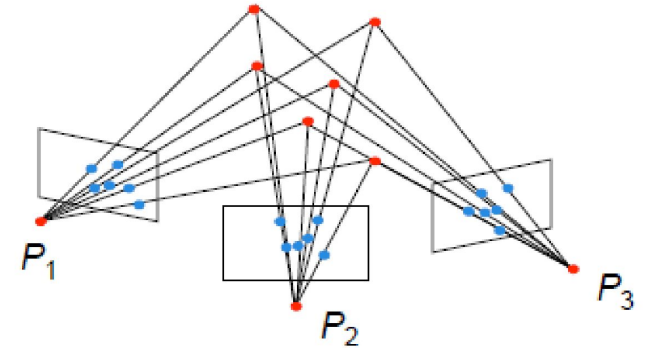


The determination of structure from motion is effectively equivalent to stereo with a single camera. The problem is that first one has to deduce the three dimensional motion of the camera between the time intervals t and $t + \delta t$.

Once this is $t + \delta t$ achieved we can then solve for the 3D positions of the matched points using the standard stereo equations.

Structure from Motion - SFM

Static scene and moving camera
-> equivalently->
rigidly moving scene and static
camera

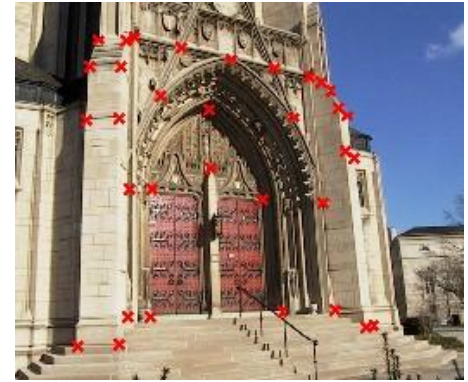
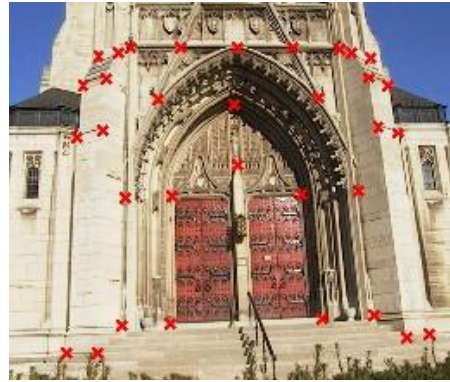
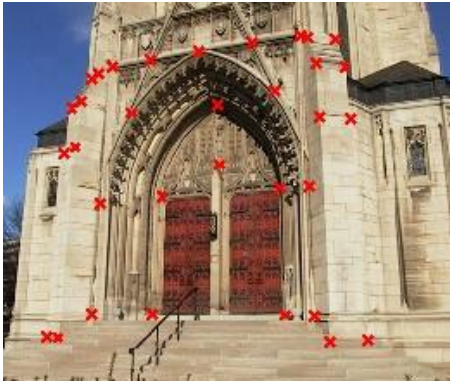


-> Limiting case of stereo with many cameras
-> Limiting case of multiview camera
calibration with unknown target

Given n points and N camera
positions, have $2nN$ equations and
 $3n+6N$ unknowns



Structure from motion



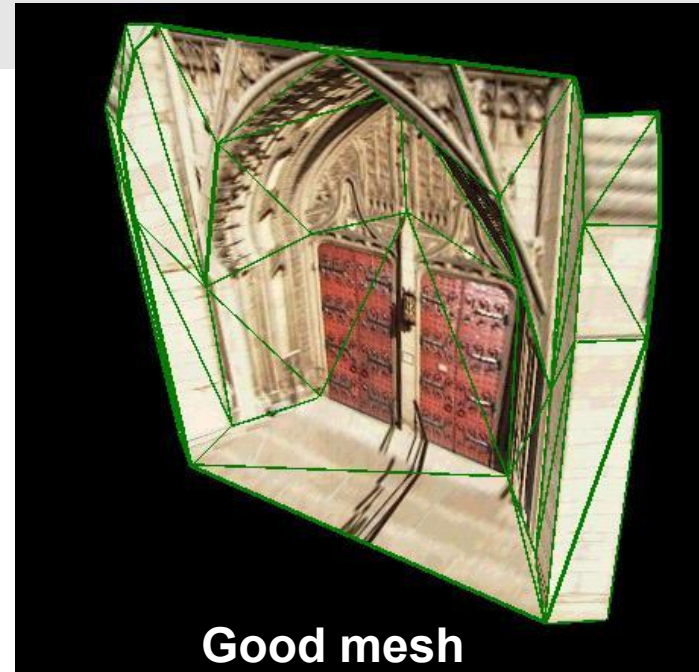
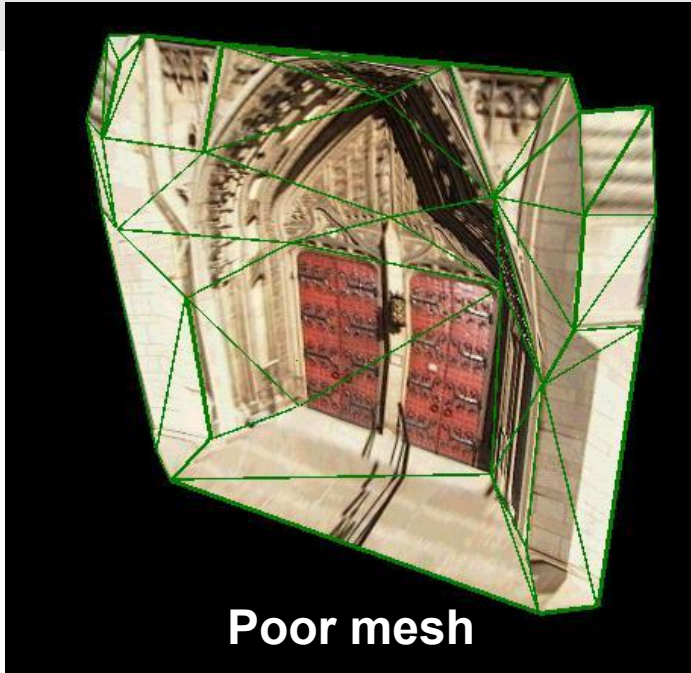
Step 1: Track Features

- Detect good features
 - corners, line segments
- Find correspondences between frames
 - Lucas & Kanade-style motion estimation
 - window-based correlation

Step 2: Estimate Motion and Structure

- Simplified projection model, e.g., [Tomasi 92]
- 2 or 3 views at a time [Hartley 00]

Structure from motion



Step 3: Refine Estimates

Step 4: Recover Surfaces

- Image-based triangulation

Example

Image sequence



[Tomasi &
Kanade]

Example

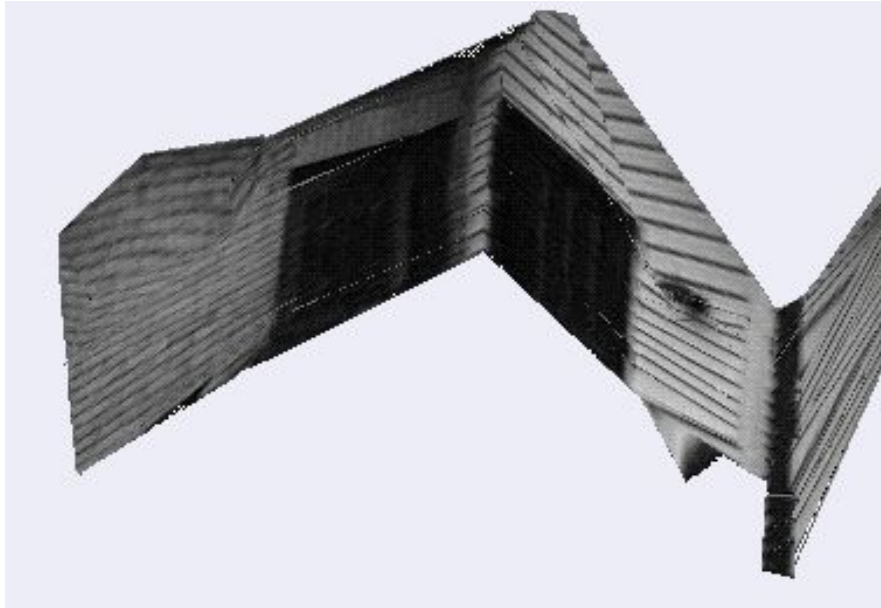
Tracked features



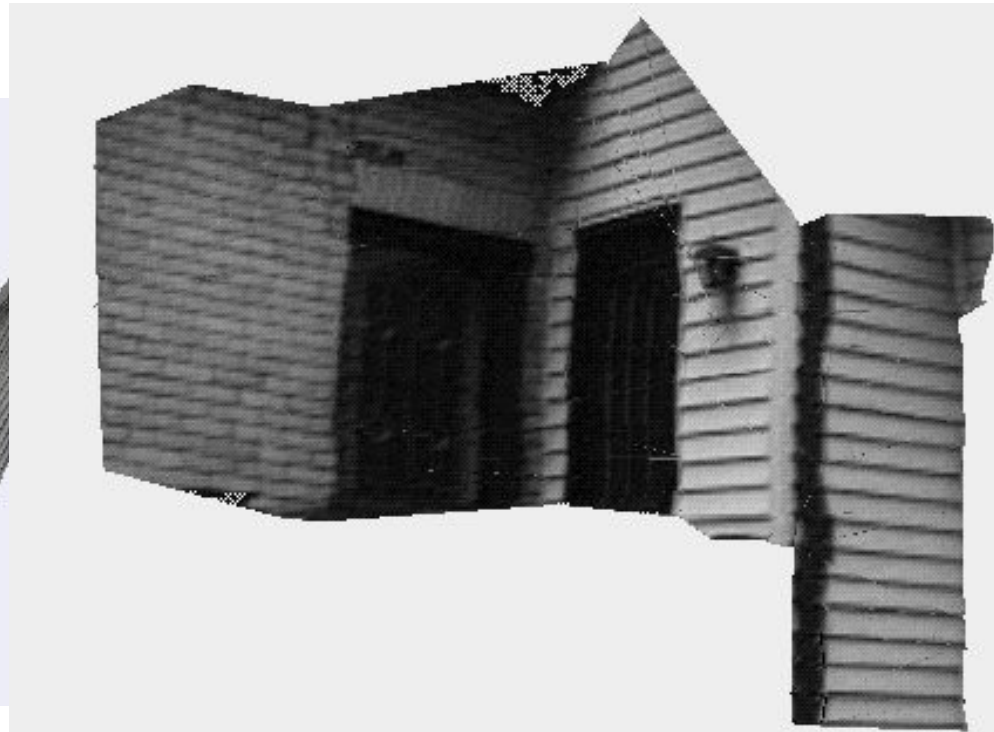
[Tomasi
&
Kanade]

Example

Reconstructed shape



Top view



Front view

[Tomasi & Kanade]

Simultaneous localisation and mapping (SLAM)

SLAM

(simultaneous localization and mapping)

SLAM is a technique used to build up a map within an unknown environment (or a known environment) while at the same time keeping track of the current location

We need:

Odometry is the use of data from motion sensors to estimate change in position over time.

Visual odometry

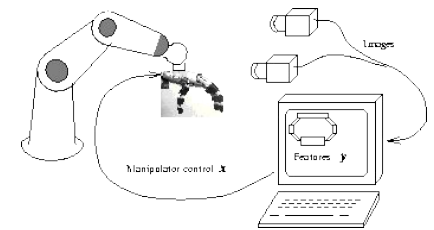
In robotics and computer vision, **visual odometry** is the process of determining the position and orientation of a device by analyzing the associated **camera images ... or laser scans...**

SLAM Applications

SLAM is central to a range of indoor, outdoor, in-air and underwater applications for both manned and **autonomous vehicles**.
SLAM is a method used for autonomous vehicles that lets you build a map and localize your vehicle in that map at the same time.

Applications requiring a low cost, portable **positioning and 3D reconstruction** device:

- **Autonomous vehicles**
- **Robotic control**
- Virtual map building (Google earth)
- Augmented reality (real-time interfaces)
- Navigation in unknown environments (i.e. Planetary landing)



Visual SLAM

Simultaneous **localisation and mapping with one or more cameras** ->
Core problem for localisation in unknown or dynamic environments.

Localisation: Estimate the 6 DoF trajectory (3D position and orientation) of a mobile camera using visual information,

Mapping: Estimate a map of the 3D environment visible to the camera(s).

In real-time i.e. $< 30\text{fps}$,

Using only vision, i.e. no gyros, accelerometers, etc.,



SLAM paradox

- SLAM is considered a fundamental problems for robots to become *truly autonomous*
- Large variety of different SLAM approaches have been developed
- The majority uses probabilistic concepts

SLAM: *robot path and map are both unknown!*

The paradox:

- In order to build a map, we must know our position
- To determine our position, we need a map
-> alternate between two steps

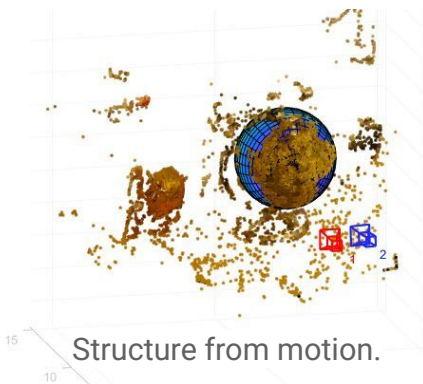
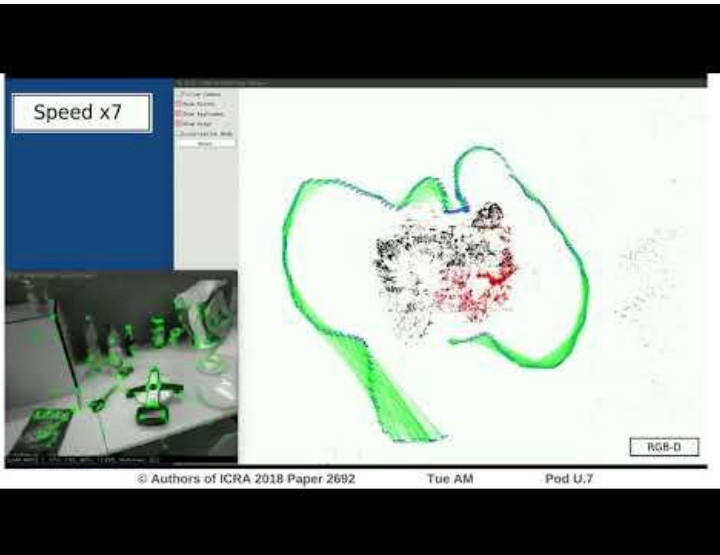
Chicken and egg problem...

Visual SLAM

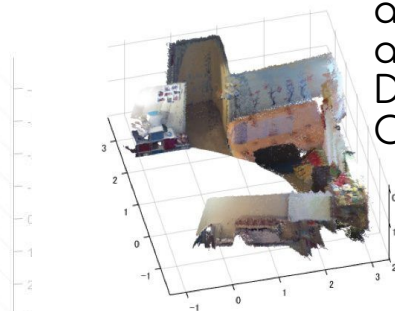
Visual SLAM (or vSLAM) uses images acquired from **cameras and other image sensors**. Visual SLAM can use simple cameras (wide angle, fish-eye, and spherical cameras), compound eye cameras (stereo and multi cameras), and RGB-D cameras (depth and ToF cameras).

Visual SLAM algorithms can be broadly classified into two categories:

- **Sparse methods** match **feature points** of images
 - Landmarks detection: Local Key points detection (Harris, Hessian, ORB...)
 - Landmarks matching
 - use algorithms such as PTAM(Parallel Tracking and Mapping) and ORB-SLAM.
- **Dense methods** use the **overall brightness of images** and use algorithms such as DTAM (Dense Tracking and Mapping in Real-Time), LSD-SLAM(Large-Scale Direct Monocular SLAM), DSO(Direct Sparse Odometry), and SVO.



Structure from motion.



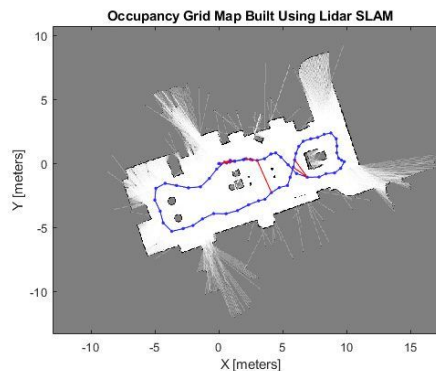
Point cloud registration for RGB-D SLAM

LiDAR (Light detection and ranging) SLAM

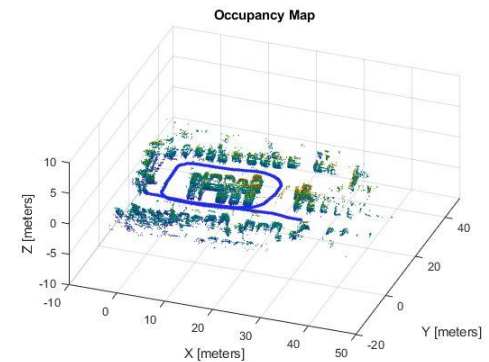
For applications such as warehouse robots, **2D lidar SLAM** is commonly used, whereas SLAM using **3-D lidar point** clouds can be used for automated driving.

The laser sensor point cloud provides high-precision distance measurements, and works very effectively for map construction with SLAM.

On the other hand, point clouds are not as finely detailed as images in terms of density and do not always provide sufficient features for matching. For example, in places where there are few obstacles, it is difficult to align the point clouds and this may result in losing track of the vehicle location.



SLAM with 2D LiDAR



SLAM with 3D LiDAR

Common Challenges with SLAM:

1. Localization errors accumulate

1. Localization **errors accumulate**, causing substantial deviation from actual values

SLAM estimates sequential movement, which include some margin of error. **The error accumulates over time, causing substantial deviation from actual values.** It can also cause map data to collapse or distort, making subsequent searches difficult.

This is called a **loop closure problem**. Pose estimation errors like these are unavoidable. It is important to detect loop closures and determine how to correct or cancel out the accumulated error.

Common Challenges with SLAM:

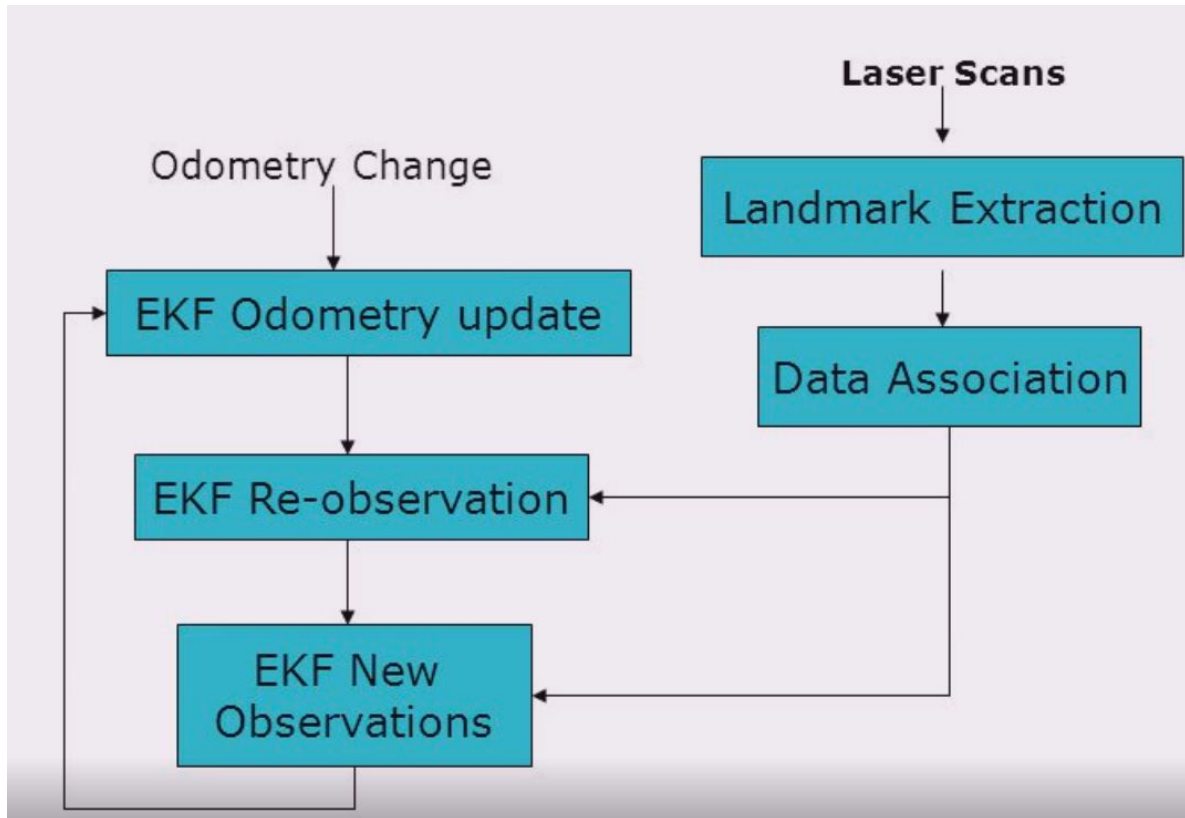
2. Localization fails and the position on the map is lost

Image and point-cloud mapping does not consider the characteristics of a robot's movement. In some cases, this approach can generate **discontinuous position** estimates.

For example, a calculation result showing that a robot moving at 1 m/s suddenly jumped forward by 10 meters. This kind of localization failure can be prevented either by using a recovery algorithm or by fusing the motion model with multiple sensors to make calculations based on the sensor data.

There are several methods for using a motion model with sensor fusion. A common method is **using Kalman filtering for localization**. Since most differential drive robots and four-wheeled vehicles generally use **nonlinear motion models, extended Kalman filters**

Overview – approach using Extended Kalman Filter



EKF: Extended Kalman Filter

Common Challenges with SLAM:

3. High computational cost

High computational cost for image processing, point cloud processing, and optimization

Problems:

- implementing SLAM on a vehicle hardware - usually performed on compact and low-energy embedded microprocessors that have limited processing power.
- image processing and point cloud matching is running at high frequency + optimization calculations such as loop closure

One countermeasure is to run different processes in parallel:

- Processes such as feature extraction, which is preprocessing of the matching process, is relatively suitable for parallelization.
- Using multicore CPUs for processing, single instruction multiple data (SIMD) calculation, and **embedded GPUs can further improve speeds in some cases.**

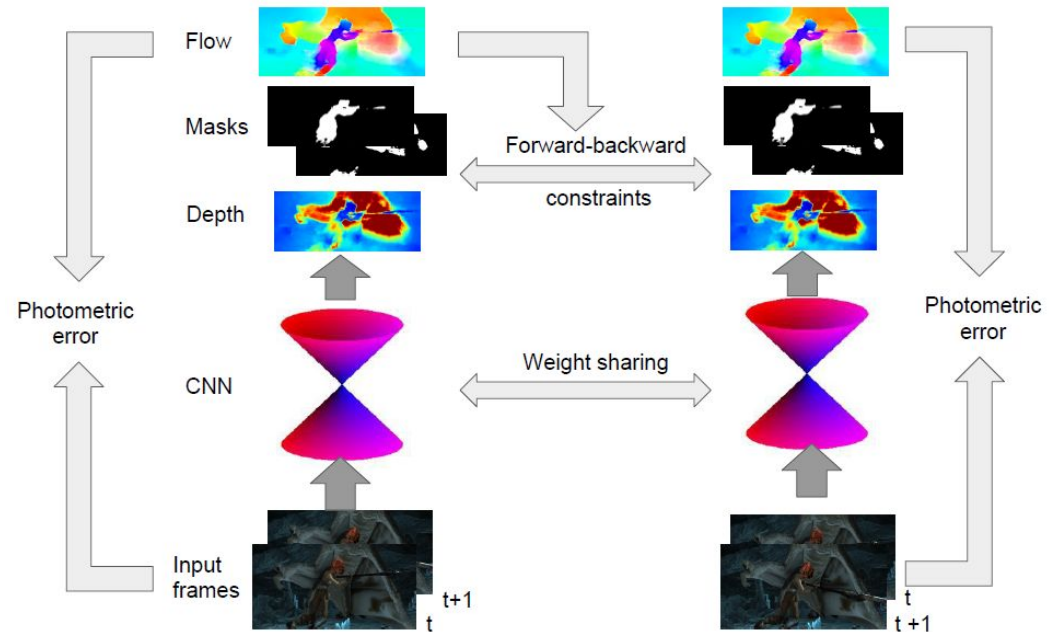
Deep learning-based 3D reconstruction

SfM - Net t: Learning of Structure and Motion from Video

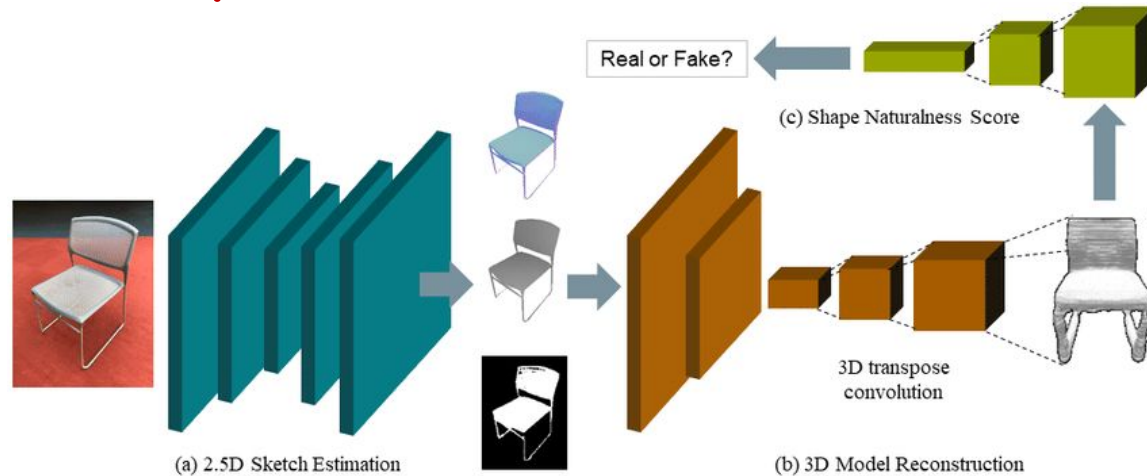
SfM-Net: Given a pair of frames as input, the model decomposes frame-to-frame pixel motion into 3D scene depth, 3D camera rotation and translation, a set of motion masks and corresponding 3D rigid rotations and translations.

It backprojects the resulting 3D scene flow into 2D optical flow and warps accordingly to match pixels from one frame to the next.

Forward backward consistency checks constrain the estimated depth.



Example of Reconstructing 3D Model from Single-View Sketch with Deep Neural Network



Single view 3D model reconstruction.

(a) Firstly, 2.5D sketches including normals, silhouettes and depths are estimated.

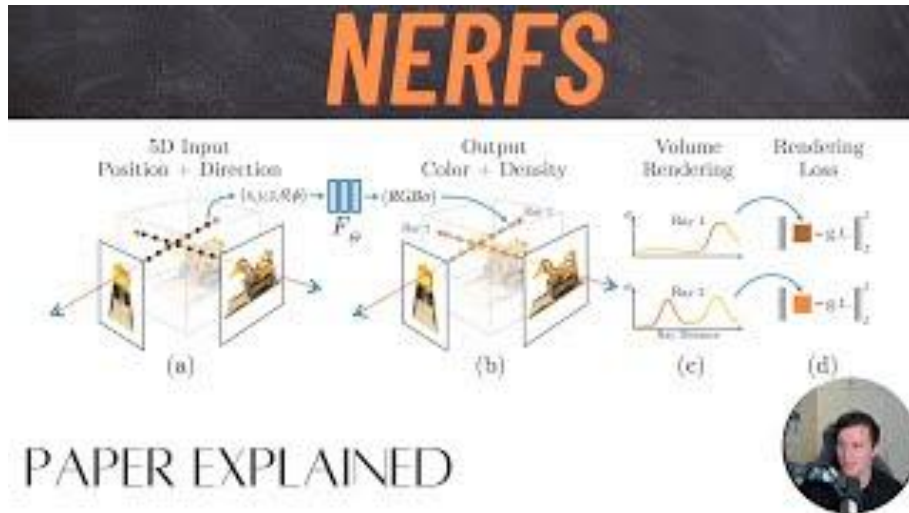
(b) Then, 3D transpose convolution is used to recover object voxels.

(c) In addition, a shape naturalness score is proposed to ensure that generated shapes are not diverged from real shapes.

NeRF

NeRFs use neural networks to represent and render realistic 3D scenes based on an input collection of 2D images.

NeRF essentially fills in the blanks, training a small neural network to reconstruct the scene by predicting the color of light radiating in any direction, from any point in 3D space.



B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "Nerf: Representing scenes as neural radiance fields for view synthesis," in European conference on computer vision. Springer, 2020, pp. 405–421

<https://blogs.nvidia.com/blog/2022/03/25/instant-nerf-research-3d-ai/>

